

Contribution Validation Report — ONE simulation run only

Case: all contributions matched

Type: Matched validation — all deterministic effects matched (expect mismatch < 0.05 m)

Duration: 600 s
Towers: 5
Receivers: 3
Max PR meas/epoch: 15
Max EKF rows/epoch: 15
Final pos error: 4.2831 m
Pos RMS (last 20%): 4.6027 m
Mean NIS: 14.95 (E[NIS] ~ 15)

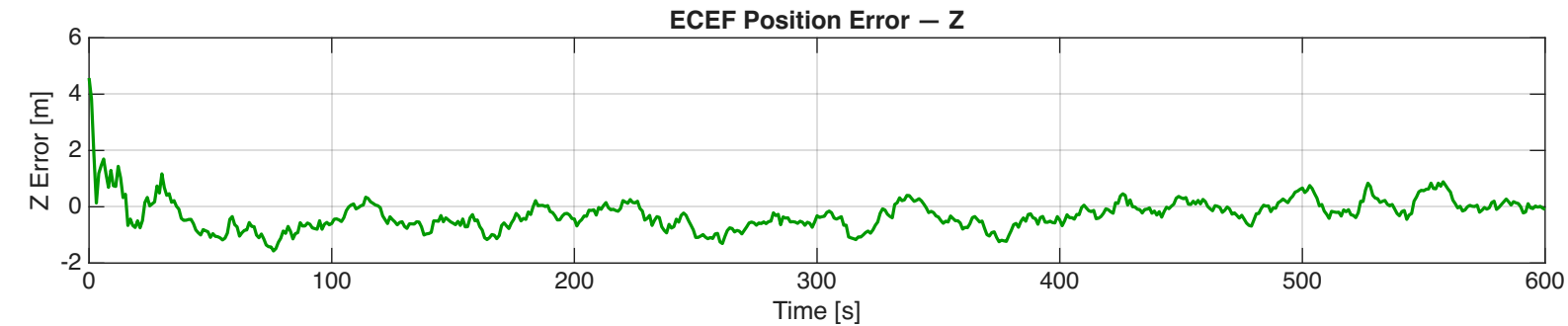
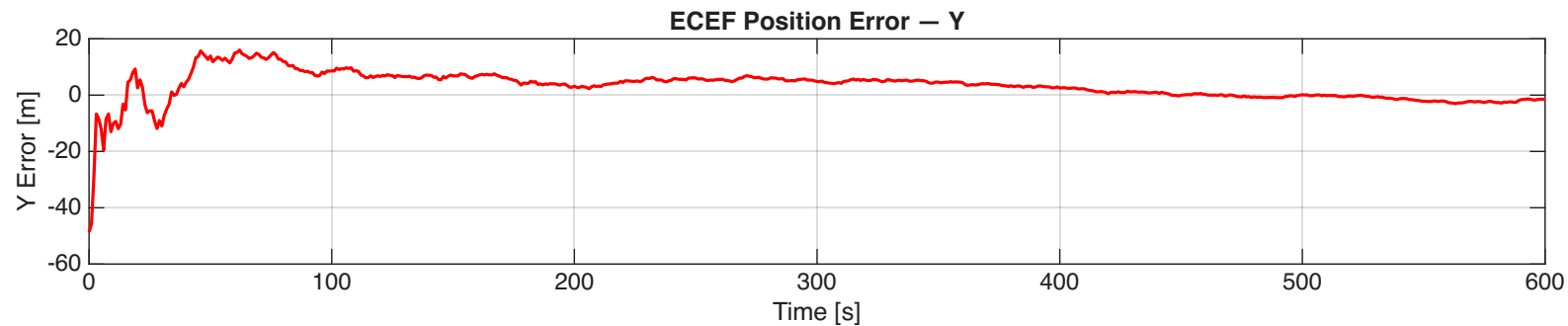
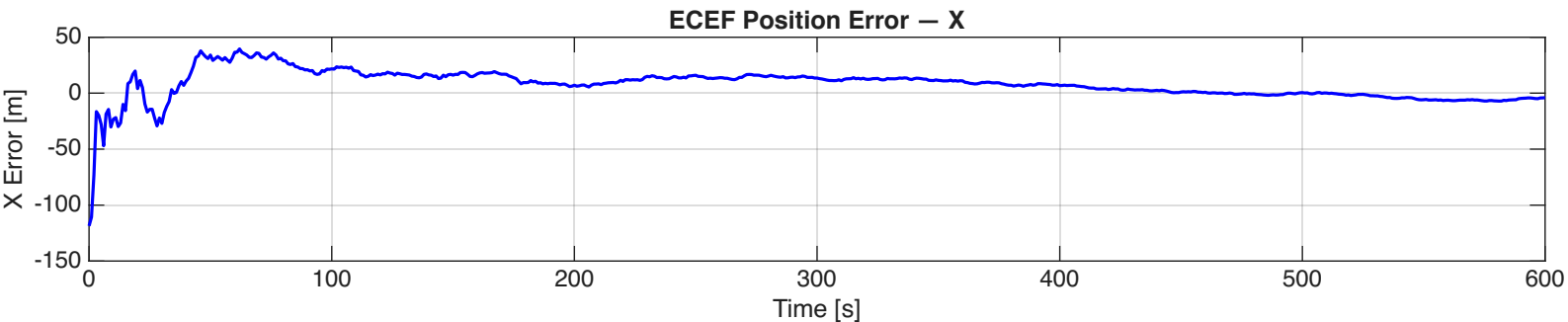
Truth effects: Sagnac, Shapiro, Trop, Iono, TowerSurvey, PCO, PCV

Model effects: Sagnac, Shapiro, Trop, Iono, TowerSurvey, PCO, PCV

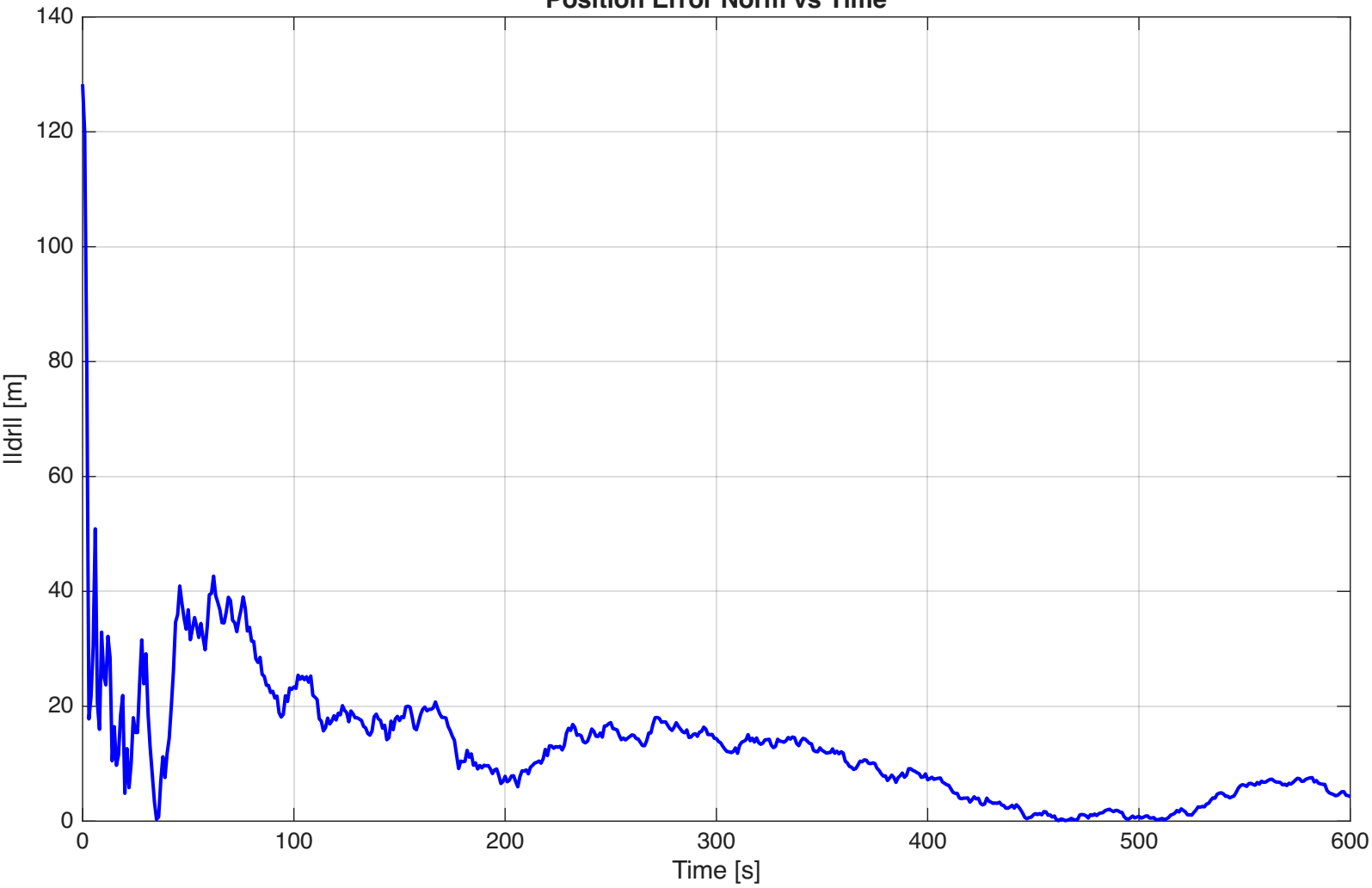
Mismatched: none

Output PDF: /Users/ludwigmatuschka/Library/CloudStorage/OneDrive-Persönlich/Dokumente/ICH/Karriere/Cranfield/IRP/Codes/IRP_simulation/oo_v1/output/effect_contribution_validation_report.pdf

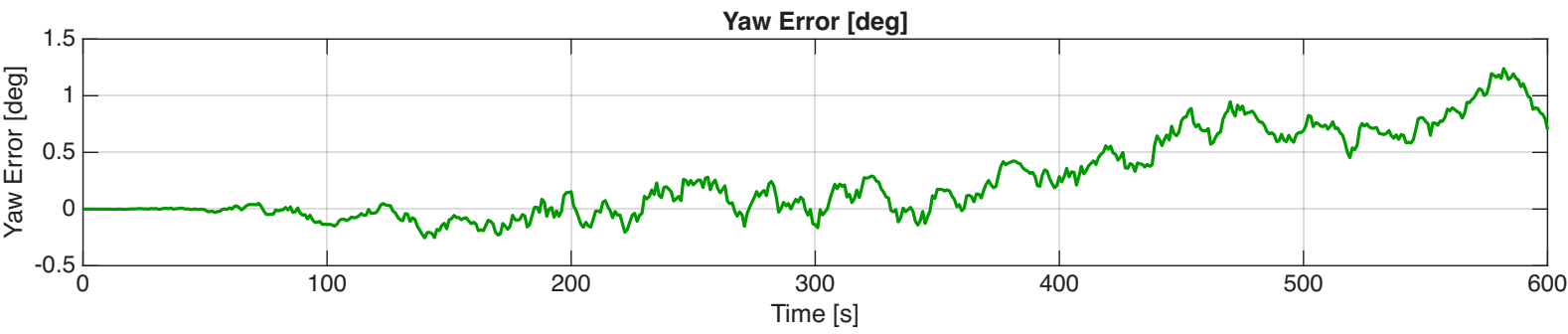
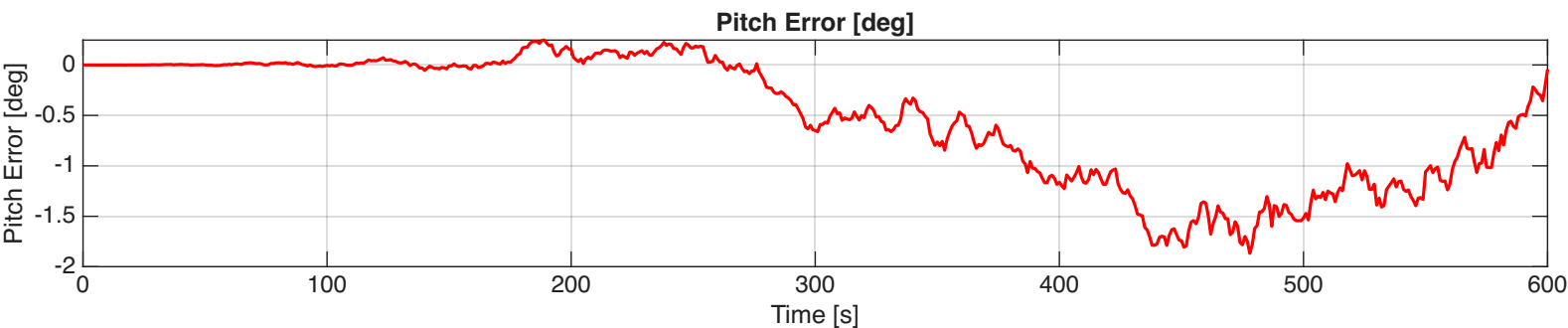
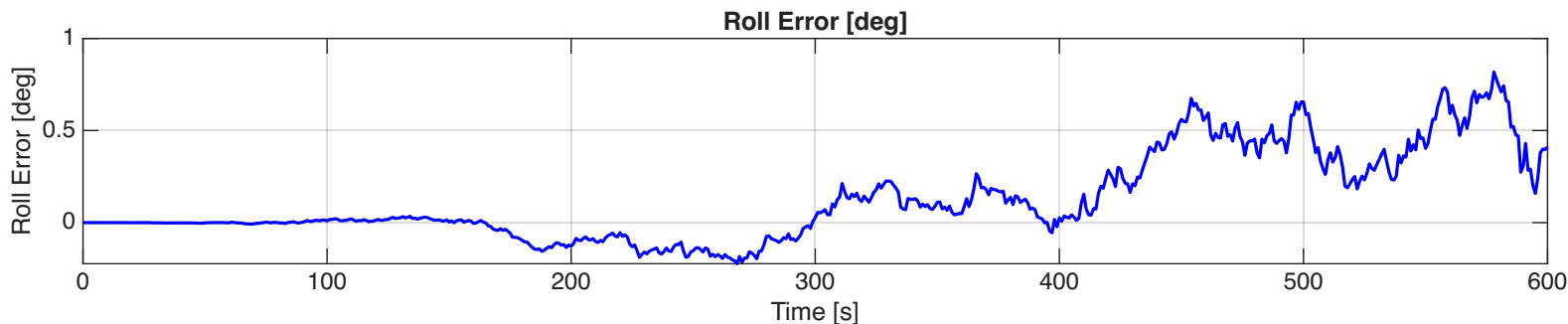
Position Error Components (ECEF)



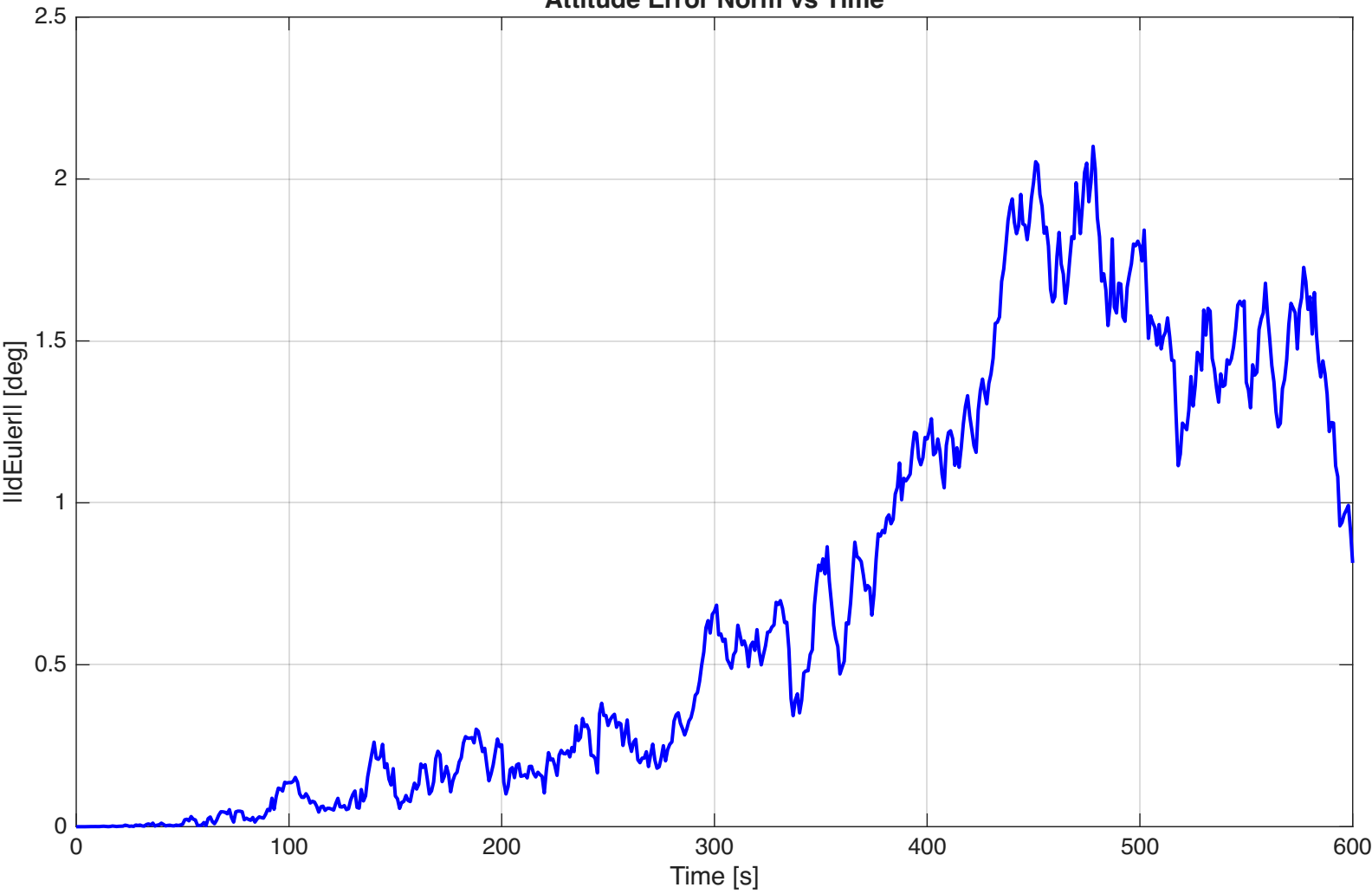
Position Error Norm vs Time



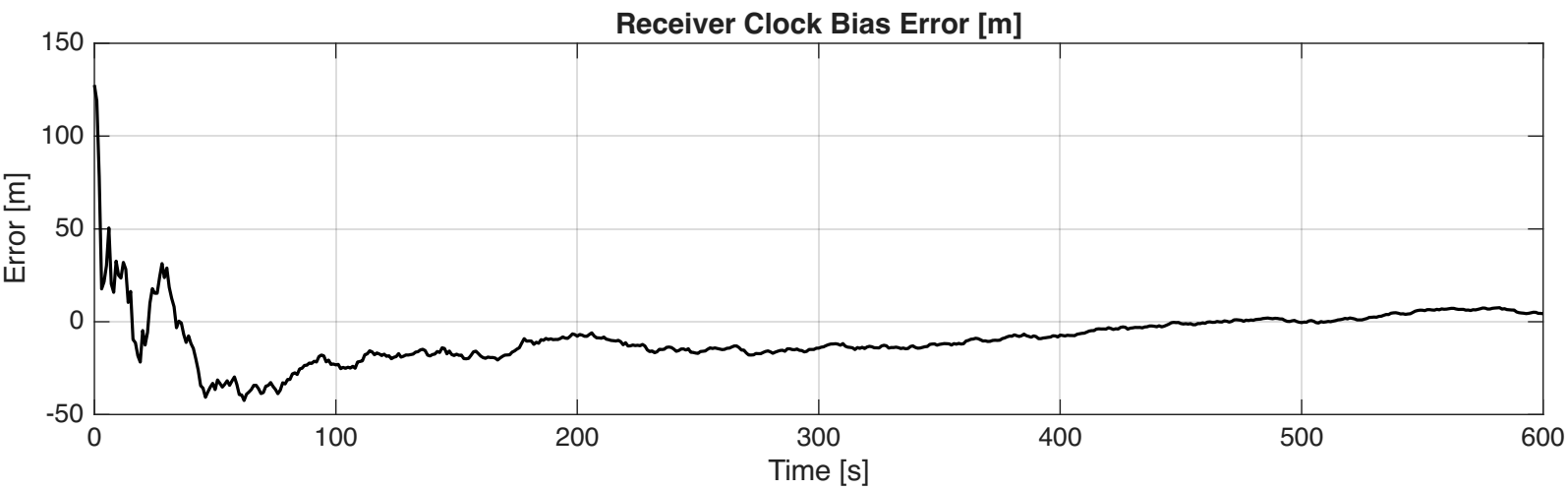
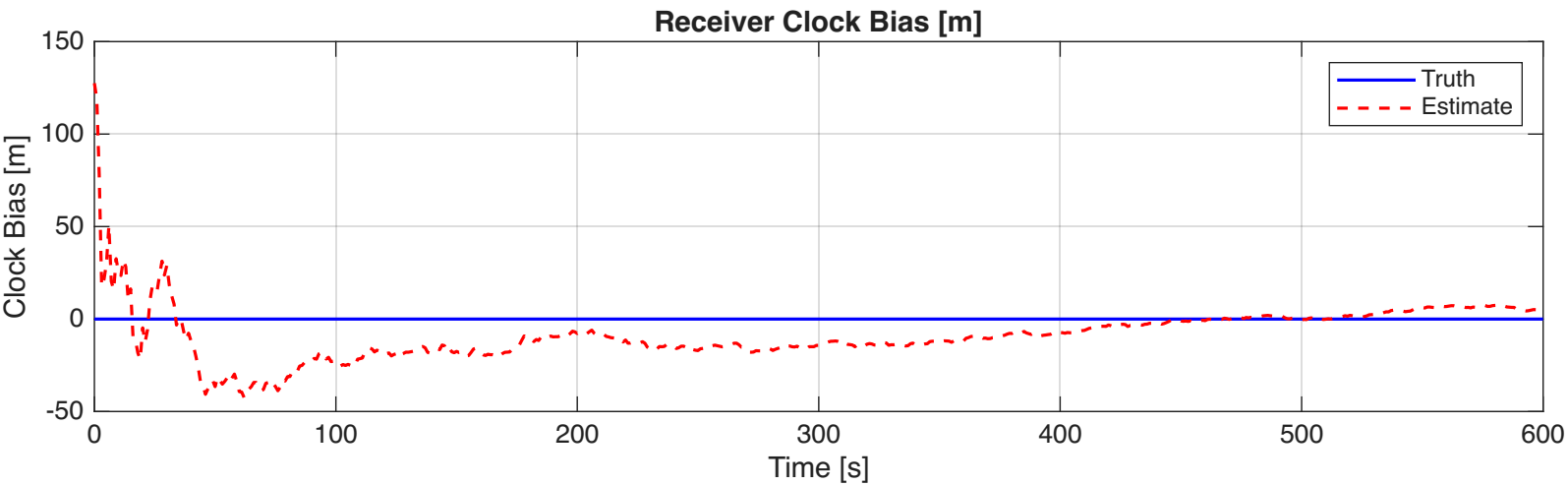
Attitude Error Components



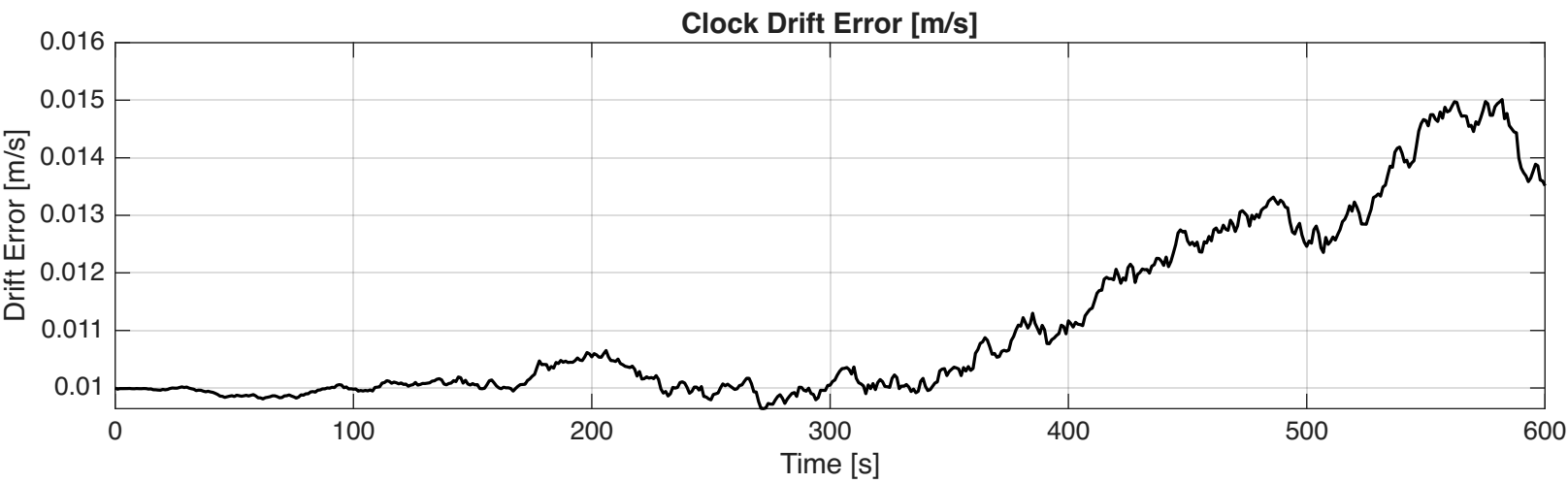
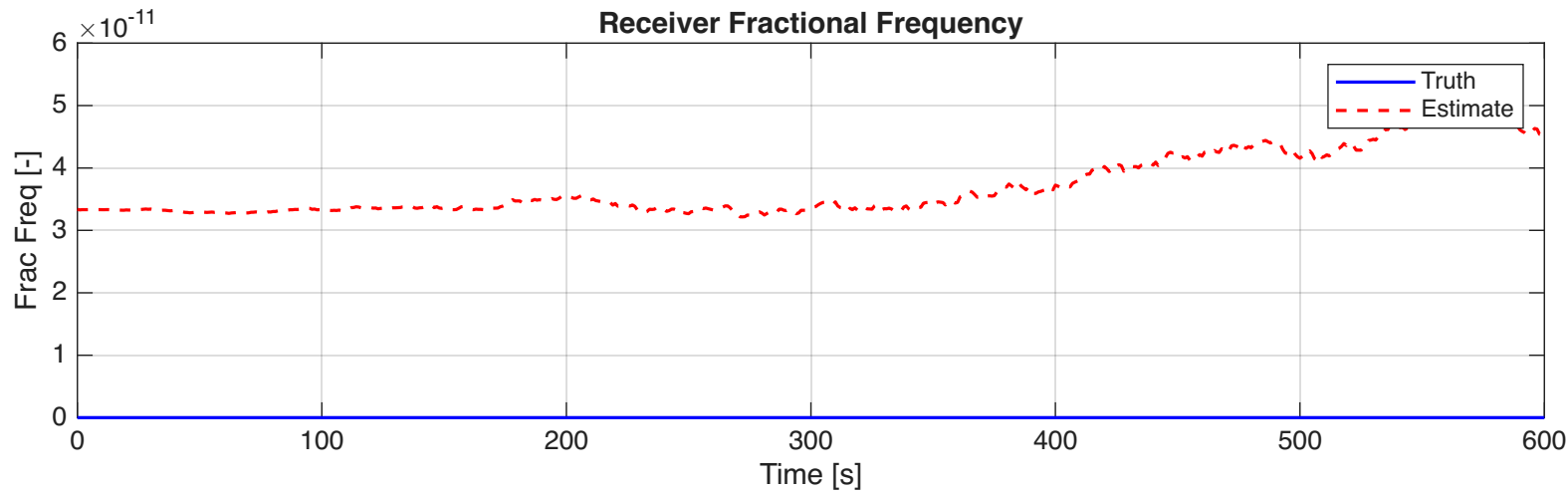
Attitude Error Norm vs Time



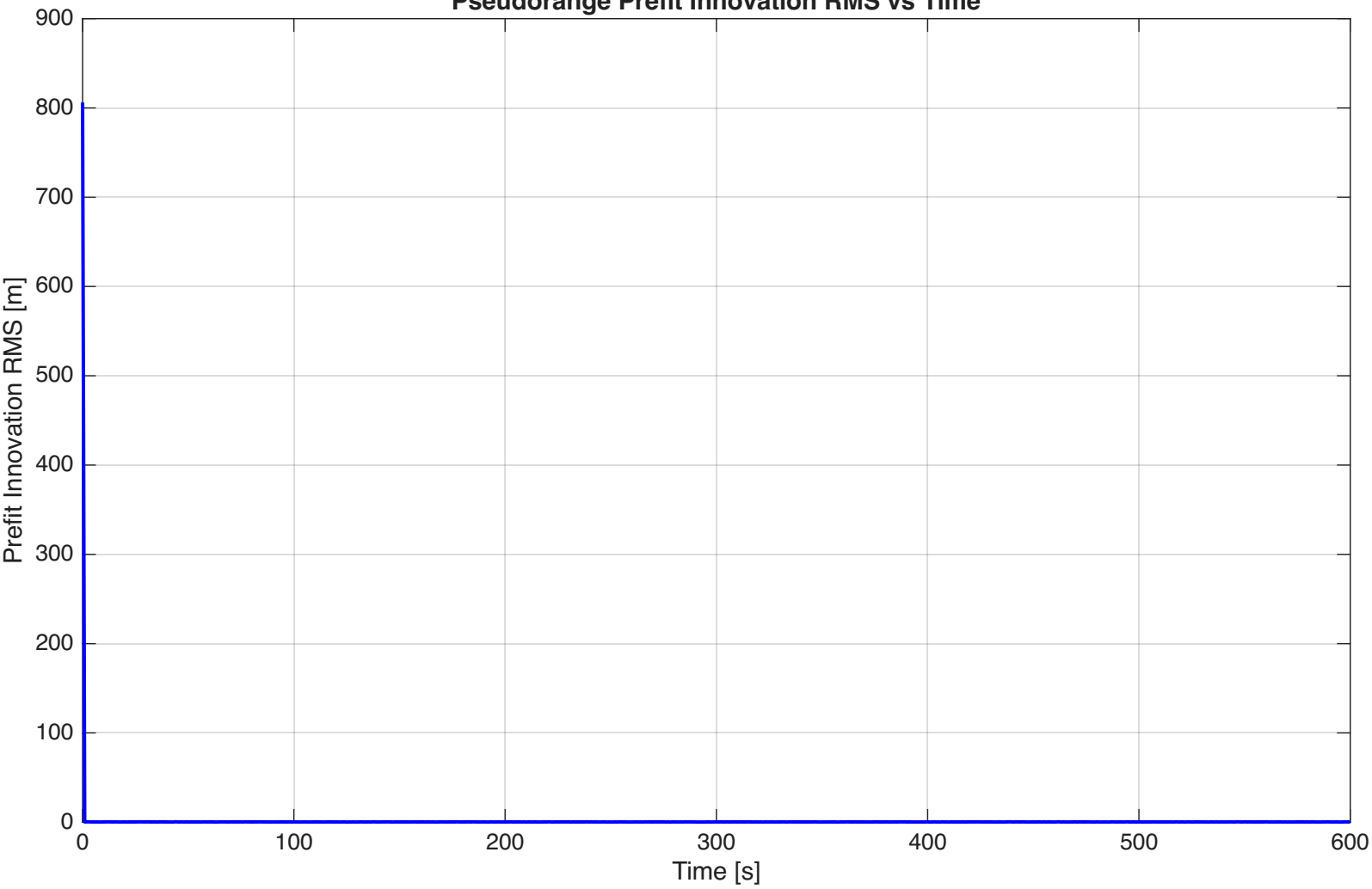
Receiver Clock Bias



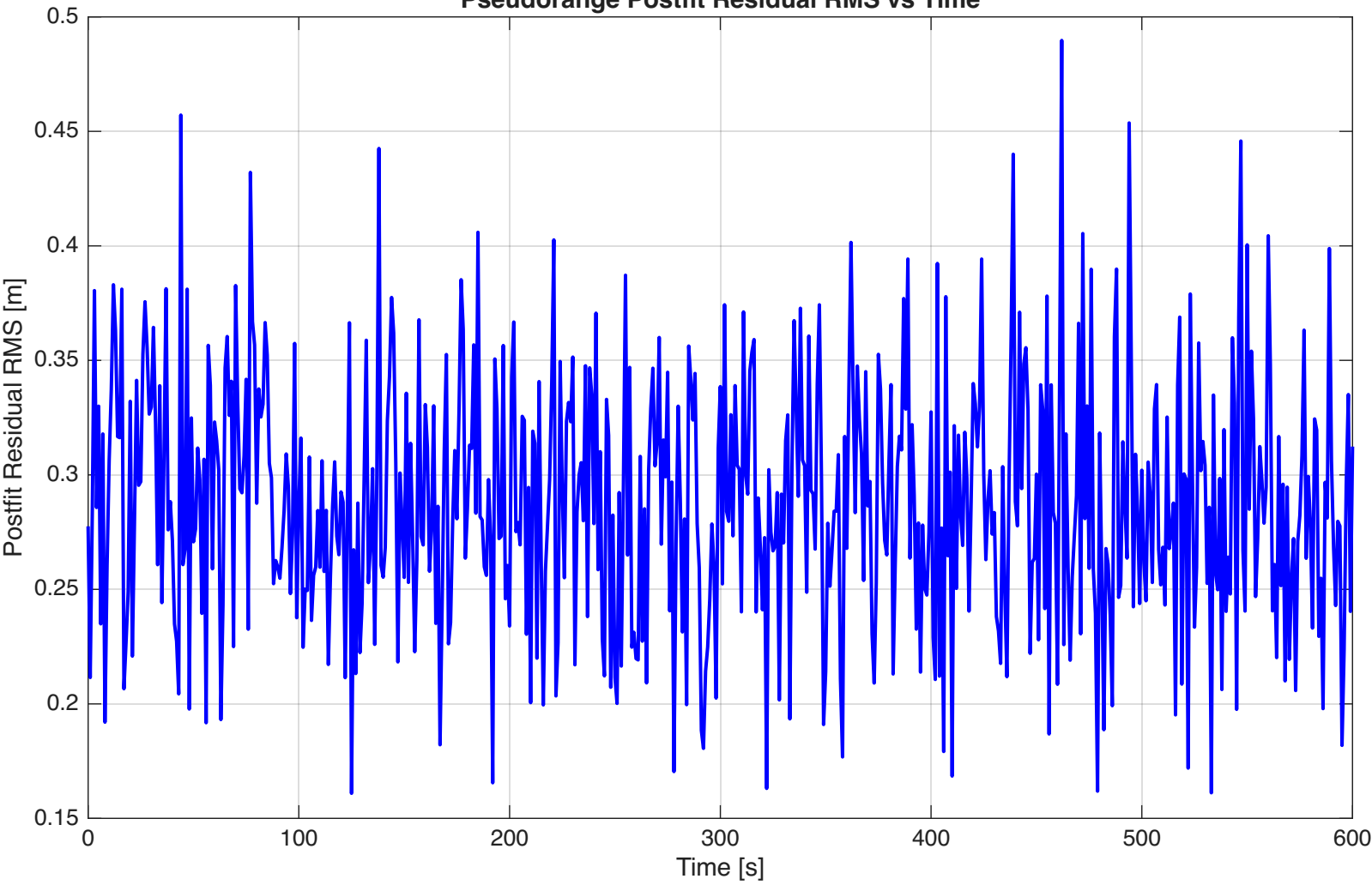
Receiver Clock Drift / Fractional Frequency



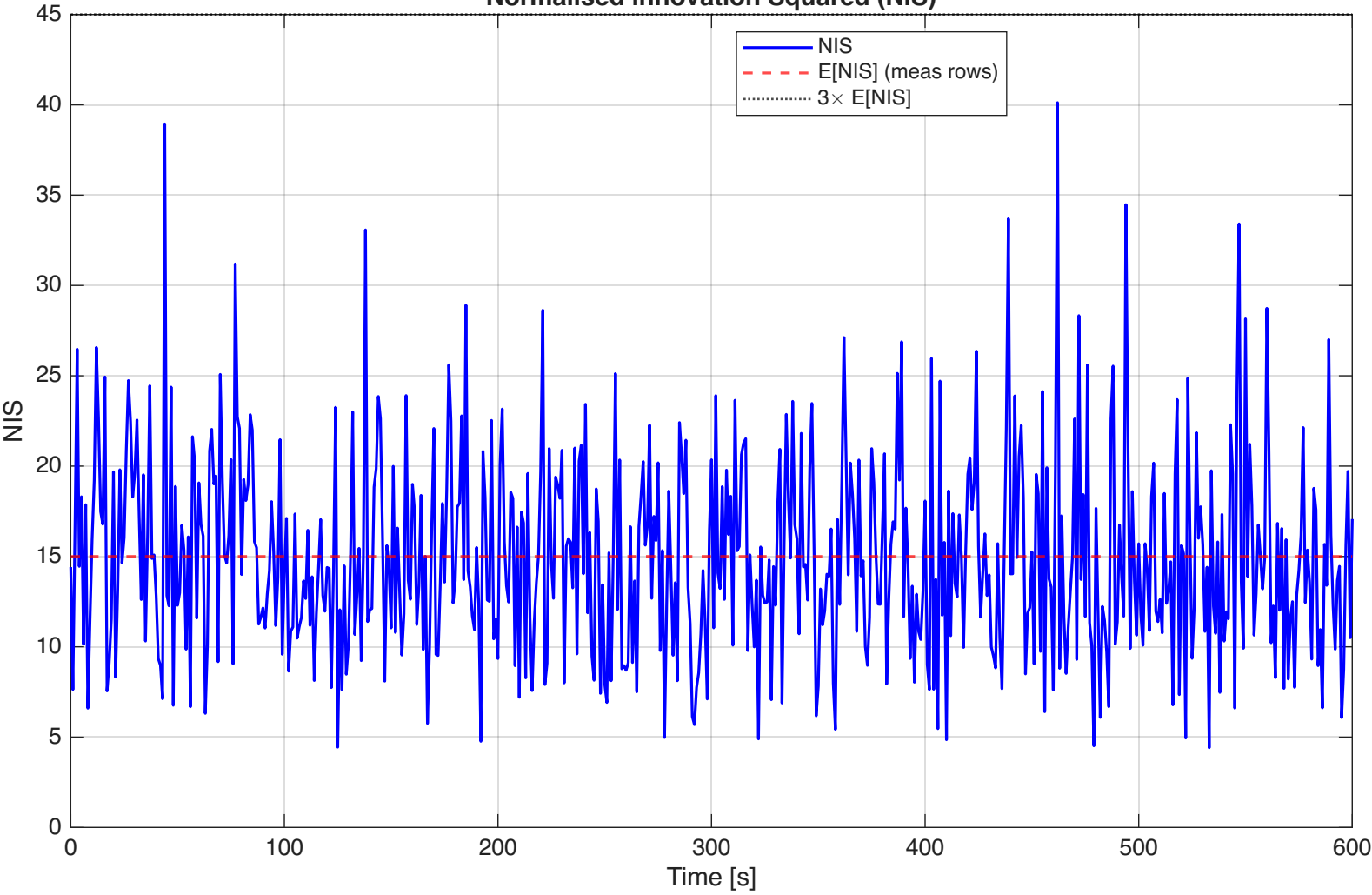
Pseudorange Prefit Innovation RMS vs Time



Pseudorange Postfit Residual RMS vs Time

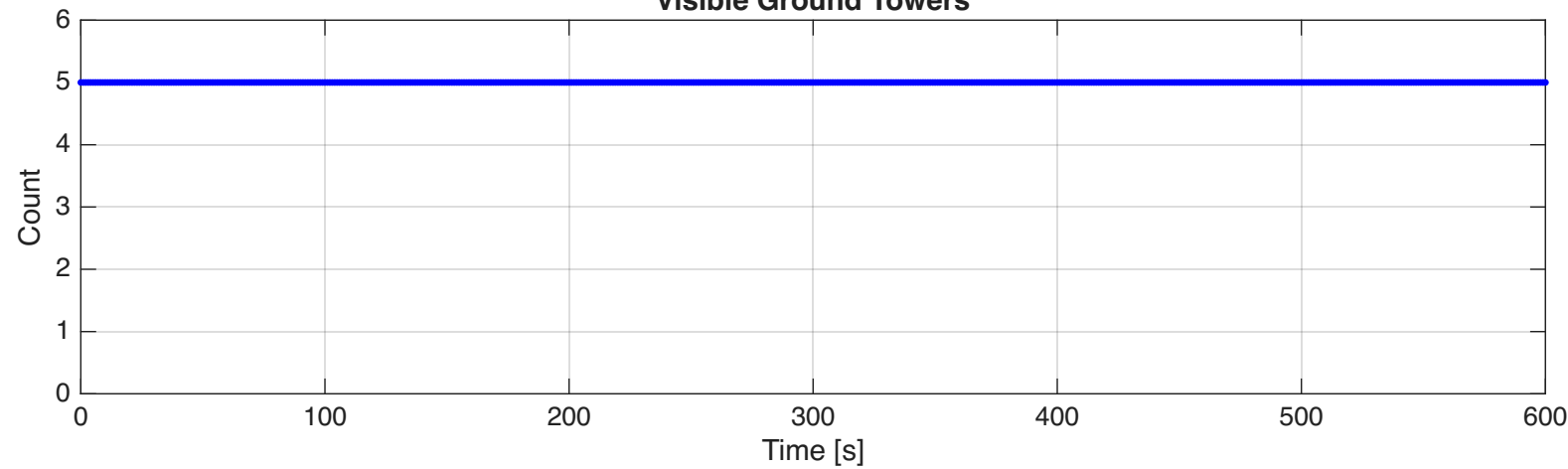


Normalised Innovation Squared (NIS)

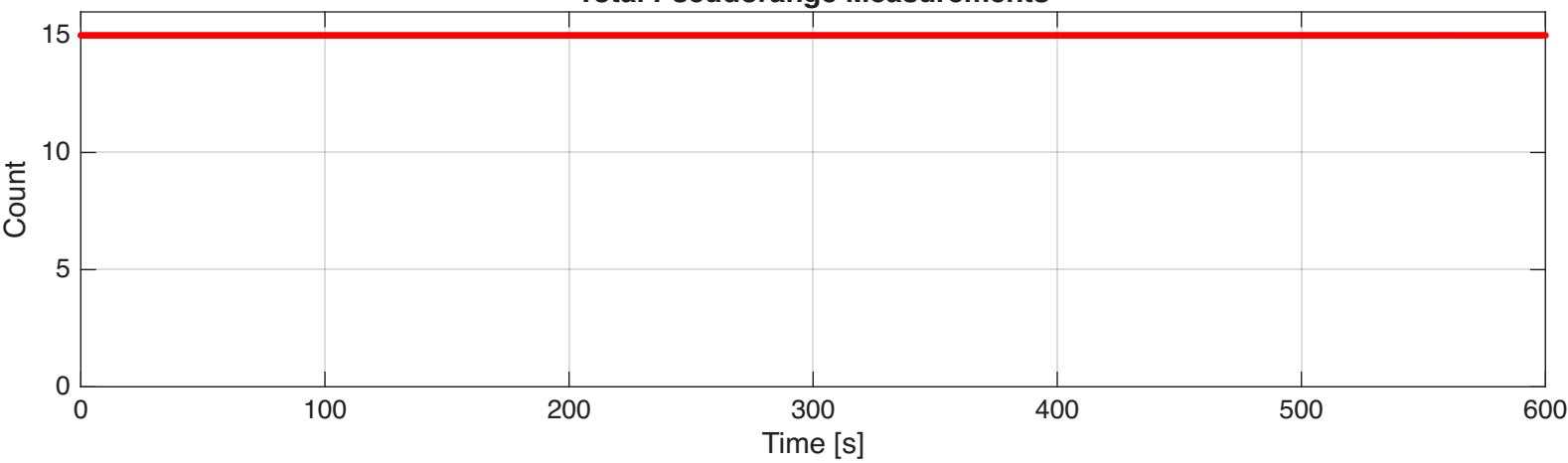


Observations per Epoch

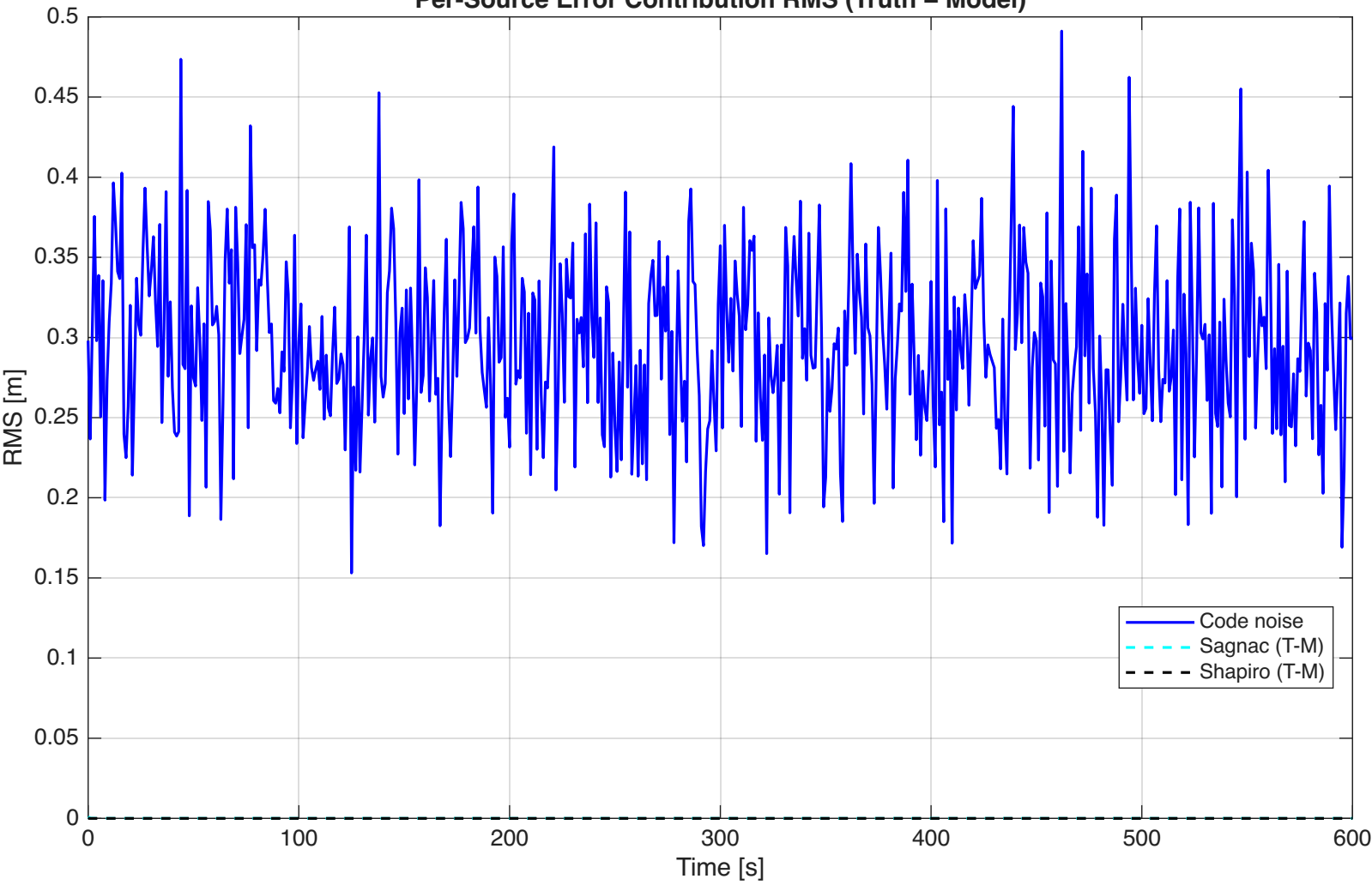
Visible Ground Towers

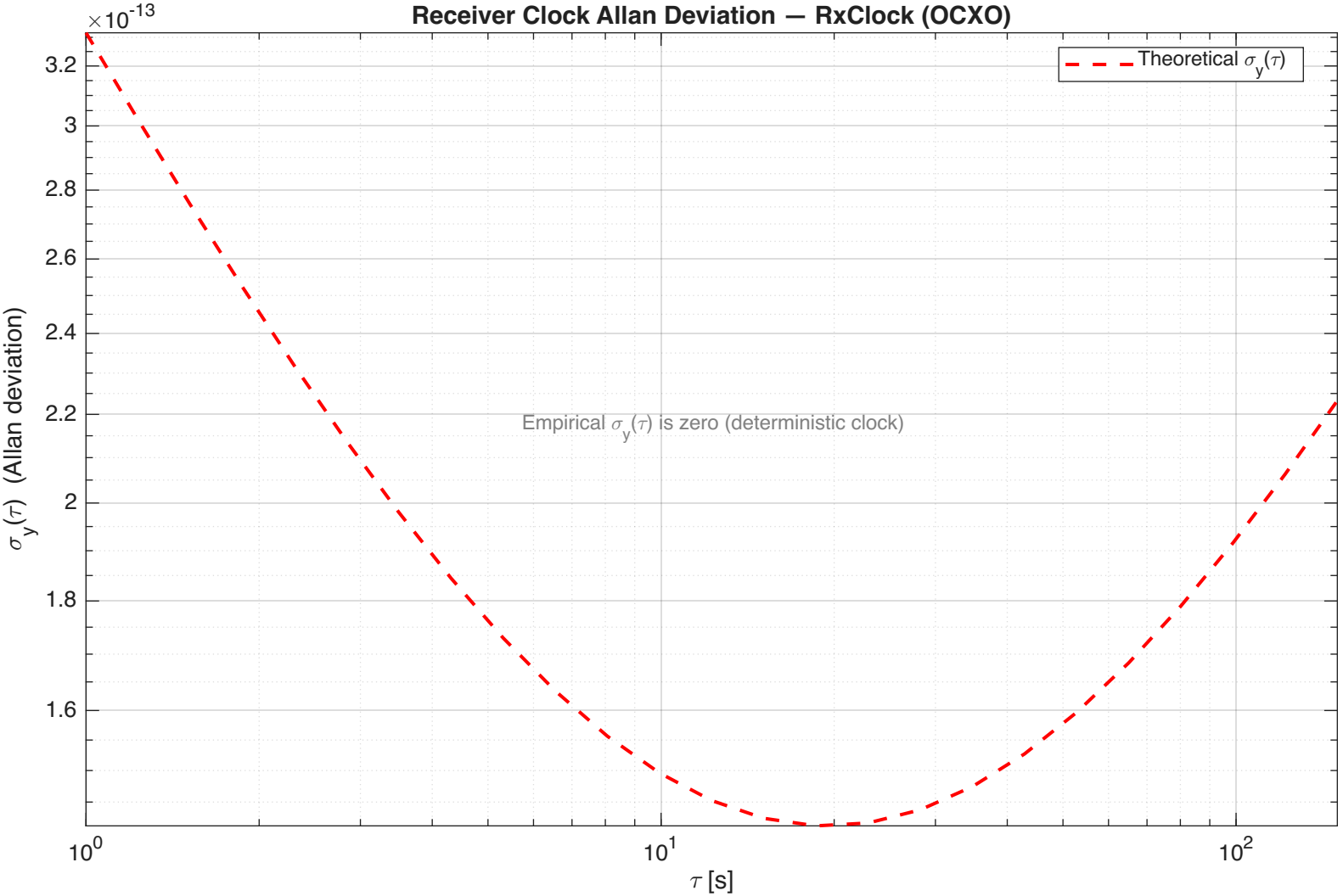


Total Pseudorange Measurements

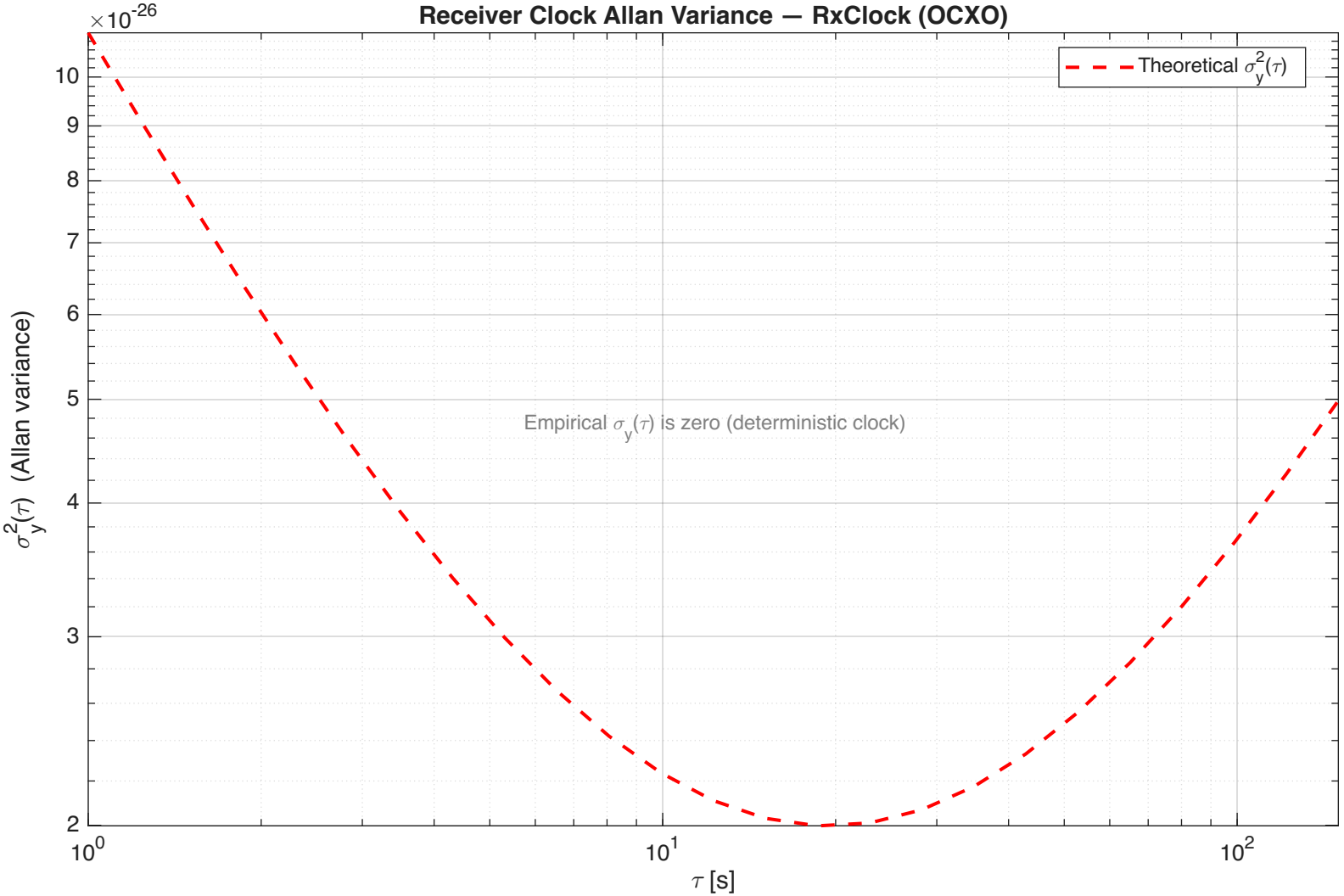


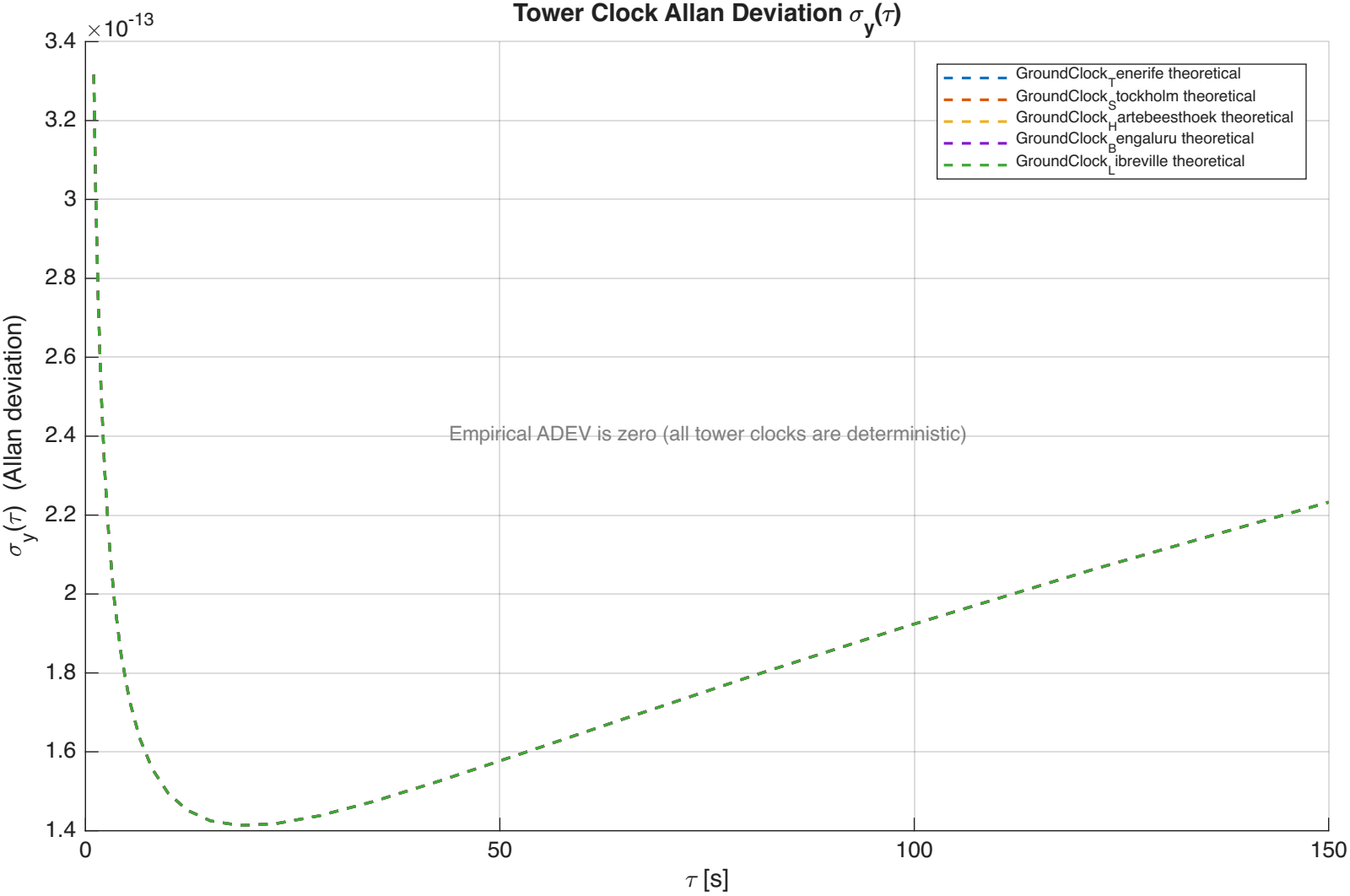
Per-Source Error Contribution RMS (Truth - Model)



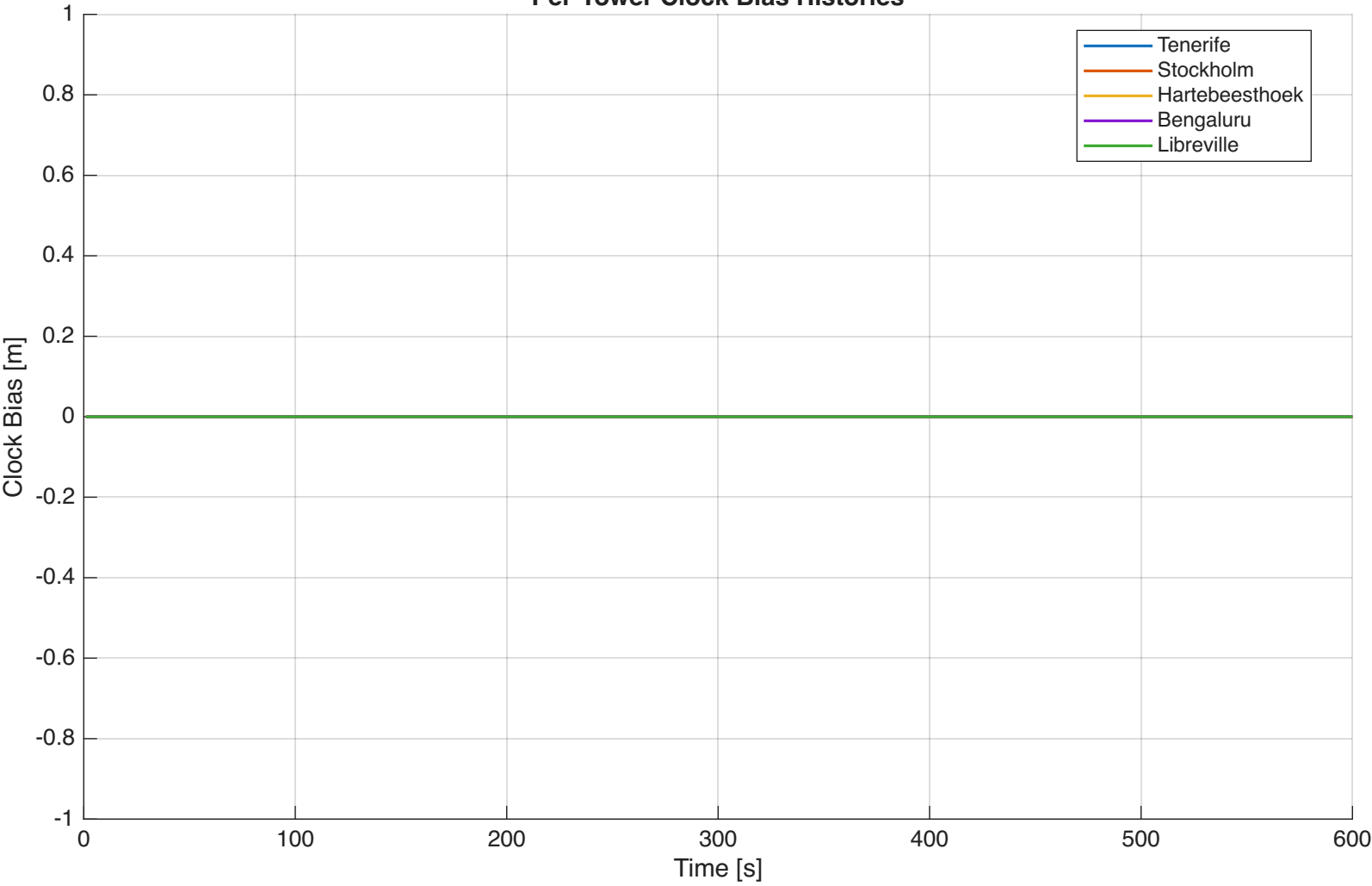


Receiver Clock Allan Variance — RxClock (OCXO)

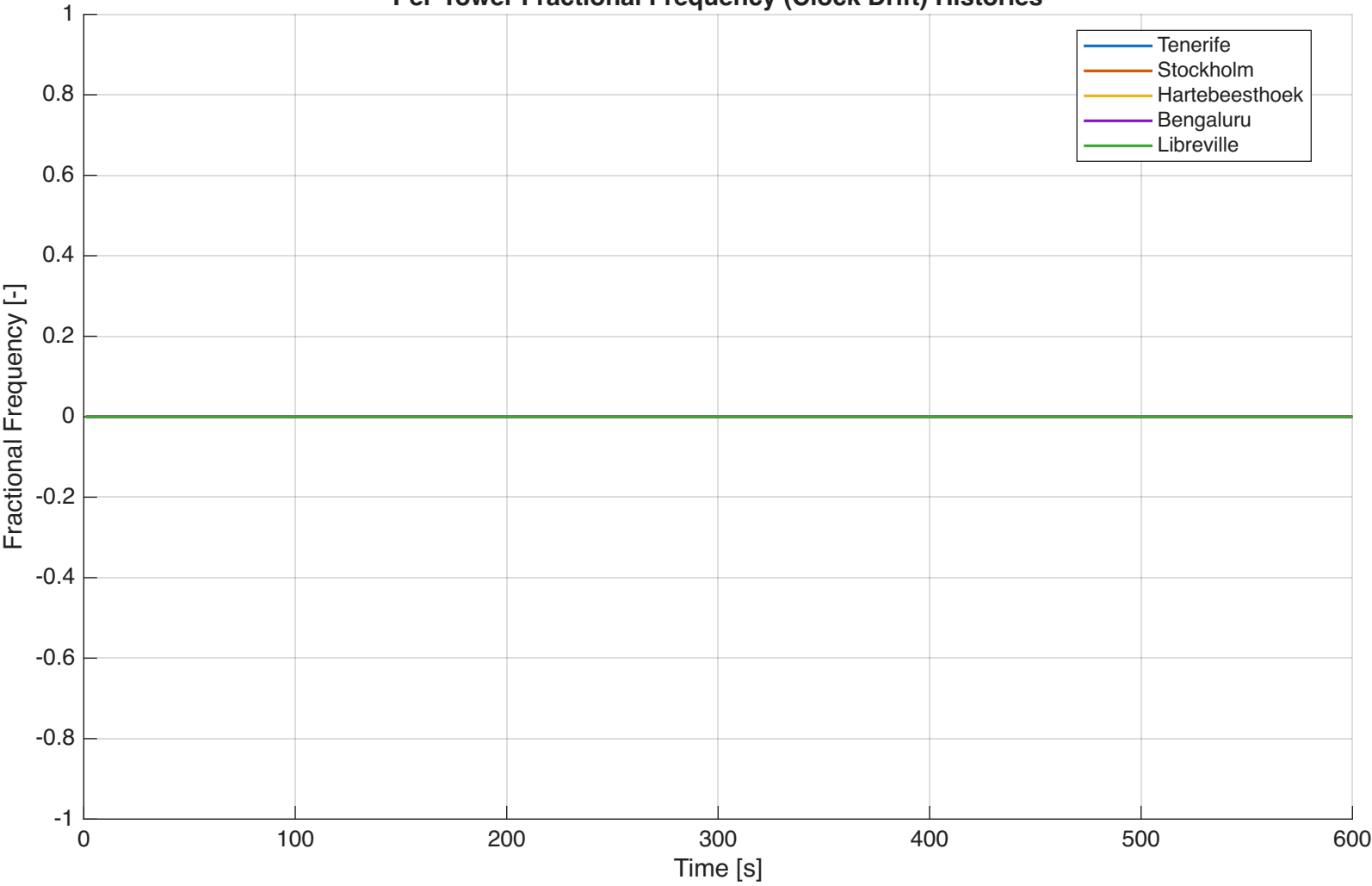


Tower Clock Allan Deviation $\sigma_y(\tau)$ 

Per-Tower Clock Bias Histories

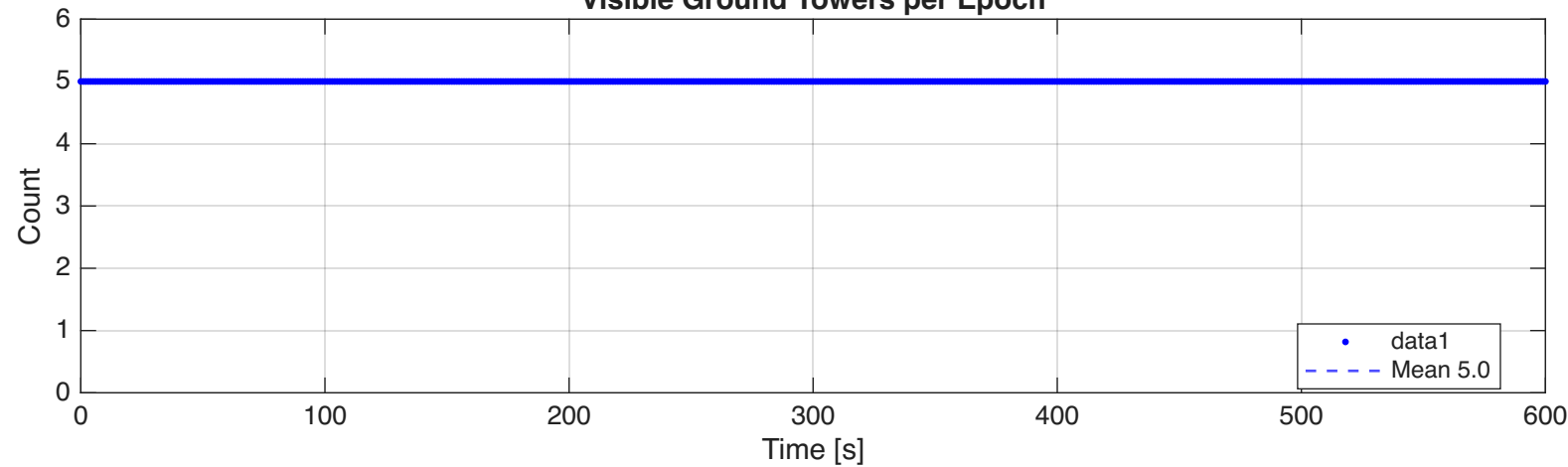


Per-Tower Fractional Frequency (Clock Drift) Histories

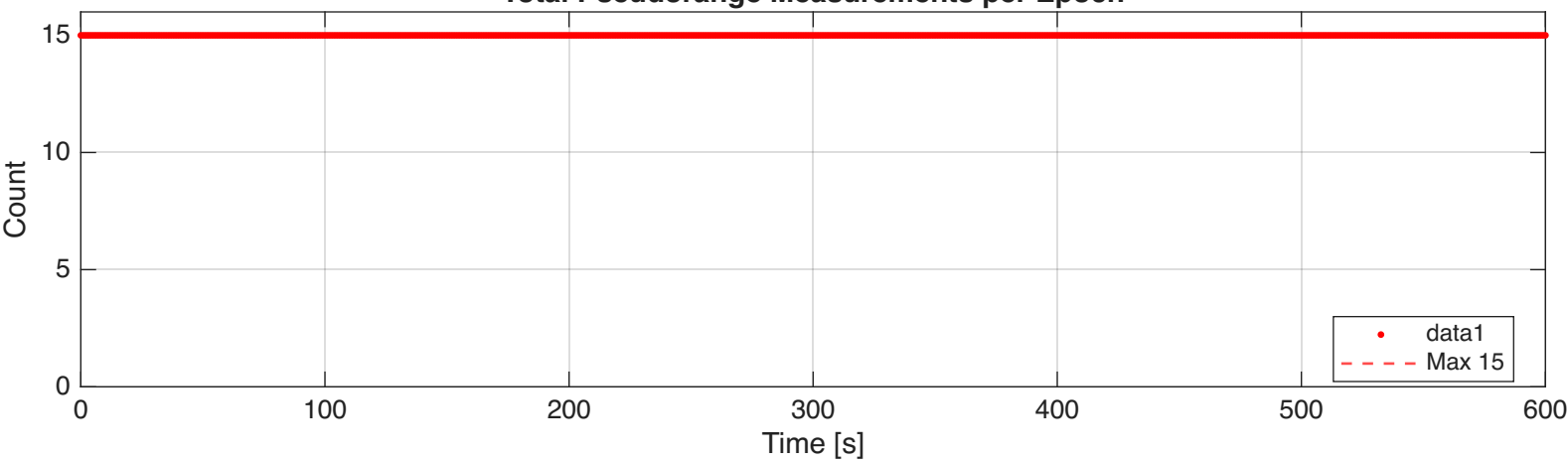


Observation Count (max meas = 15)

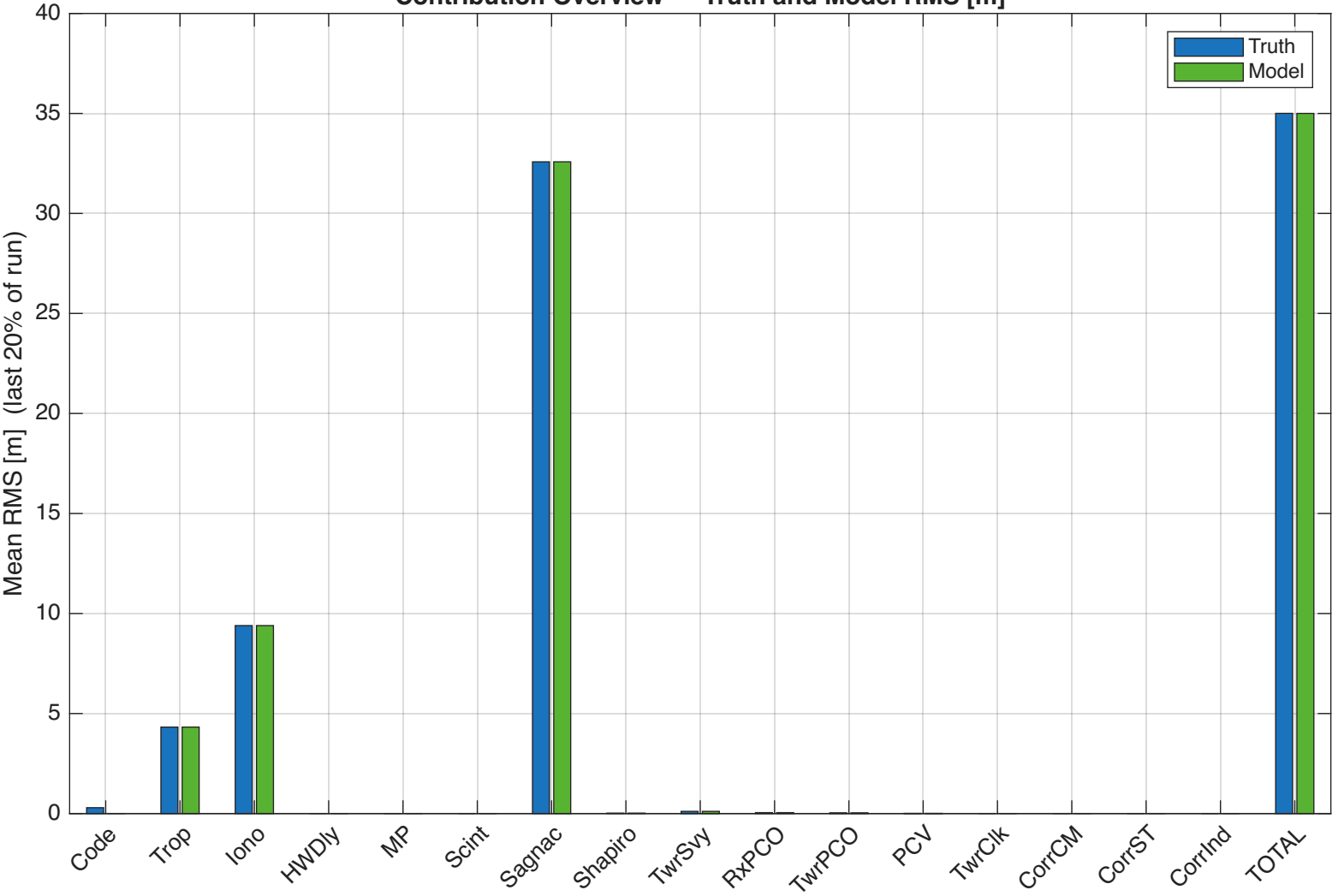
Visible Ground Towers per Epoch



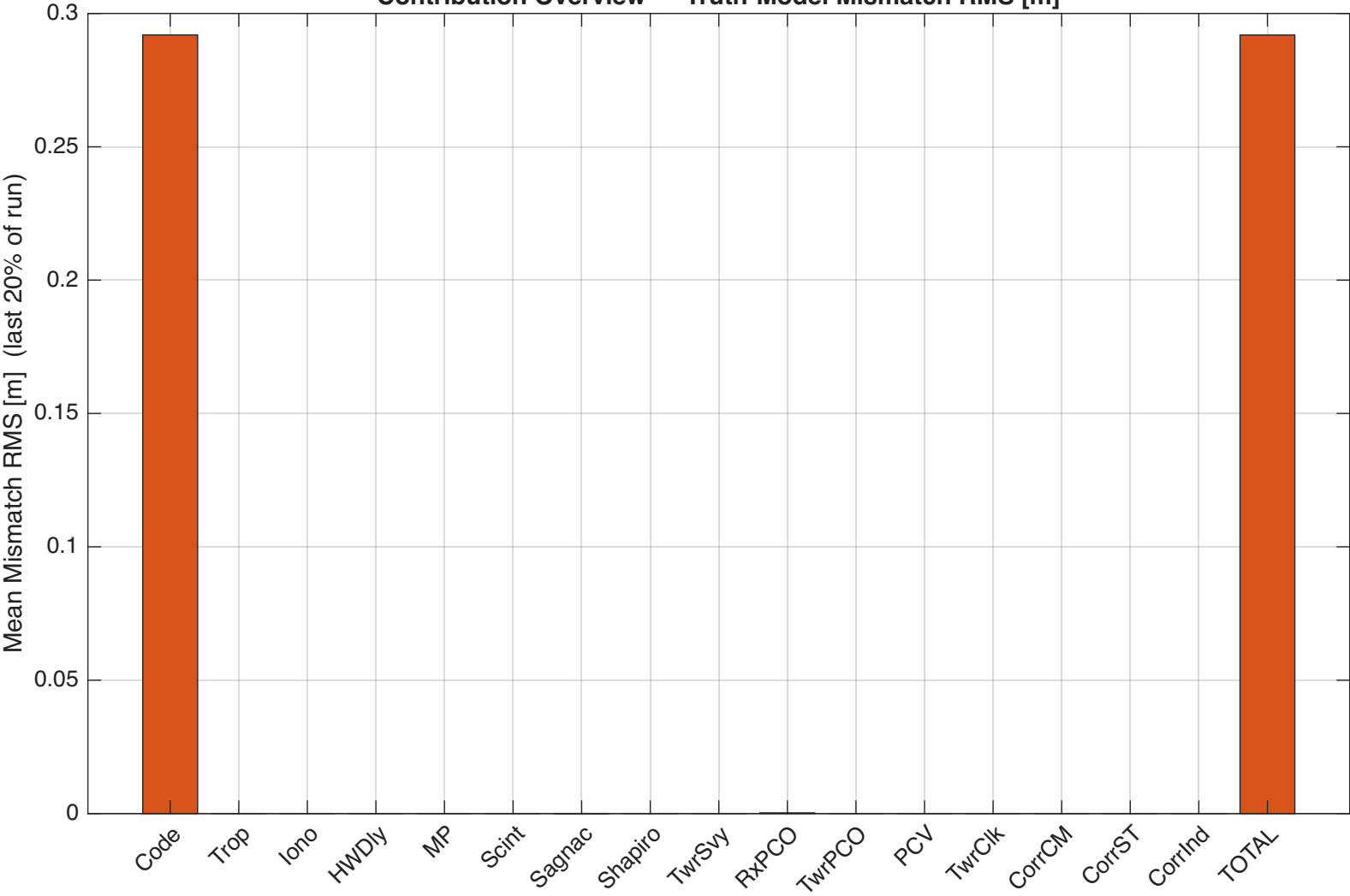
Total Pseudorange Measurements per Epoch

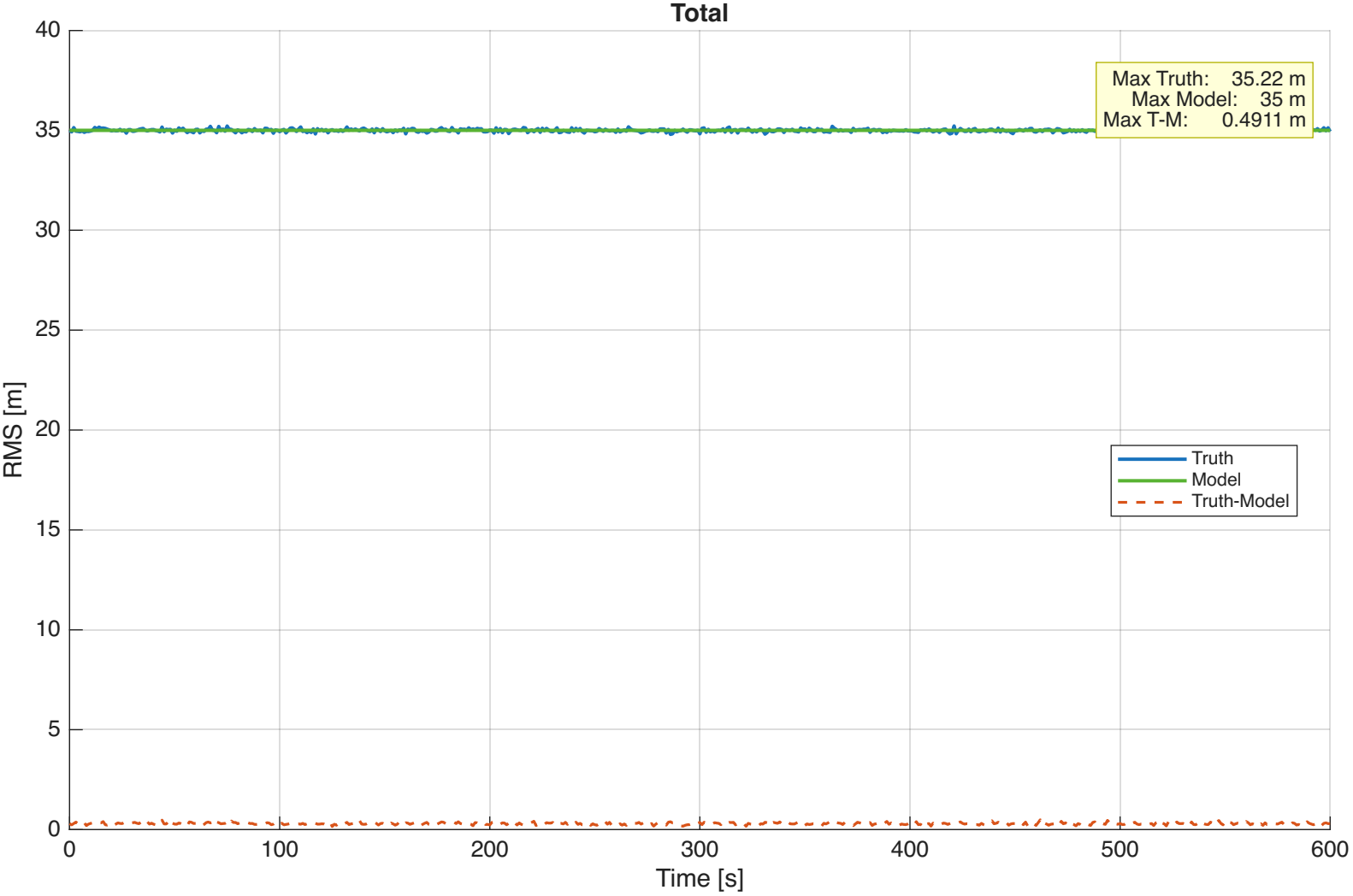


Contribution Overview — Truth and Model RMS [m]

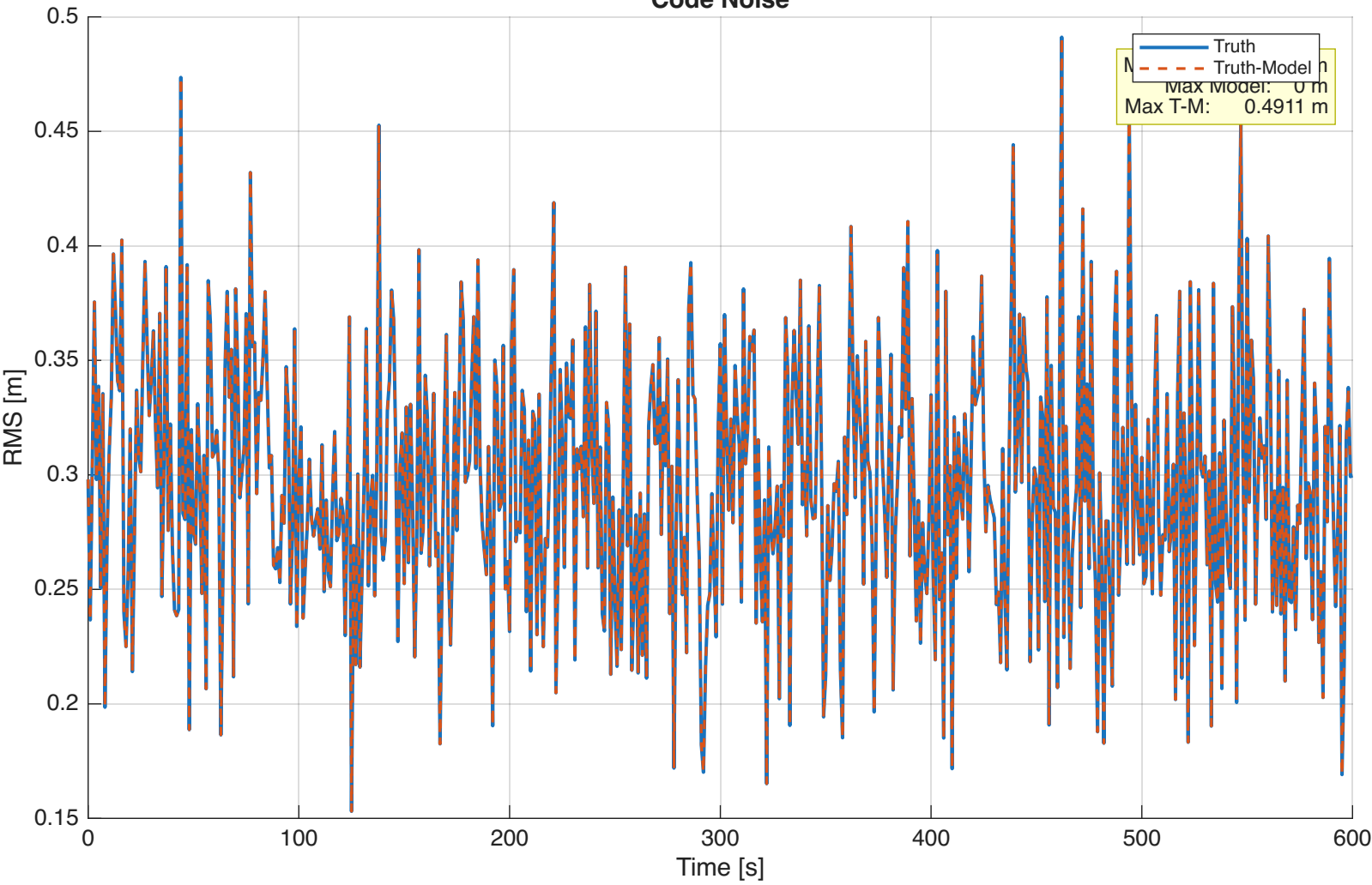


Contribution Overview — Truth-Model Mismatch RMS [m]

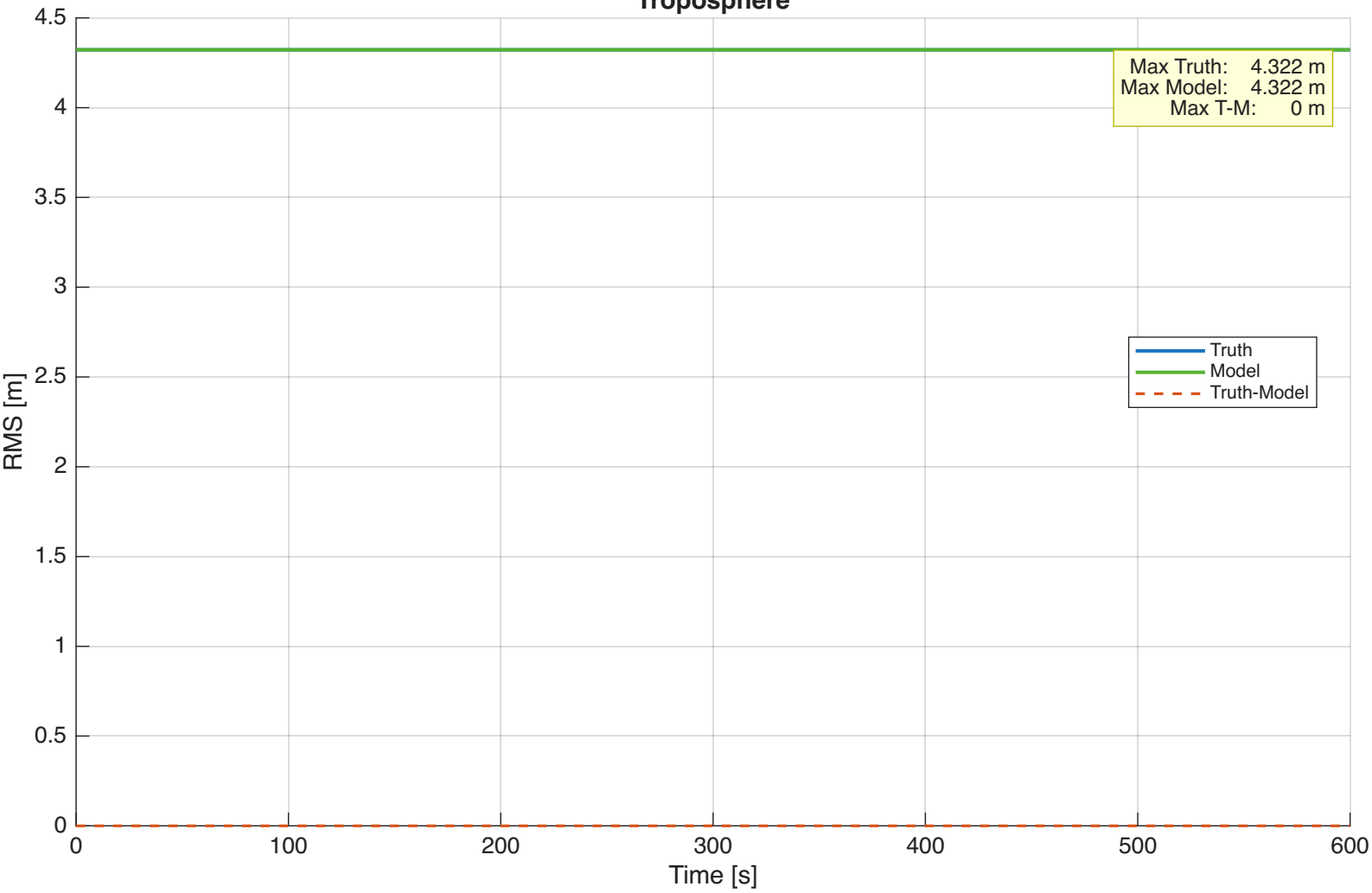




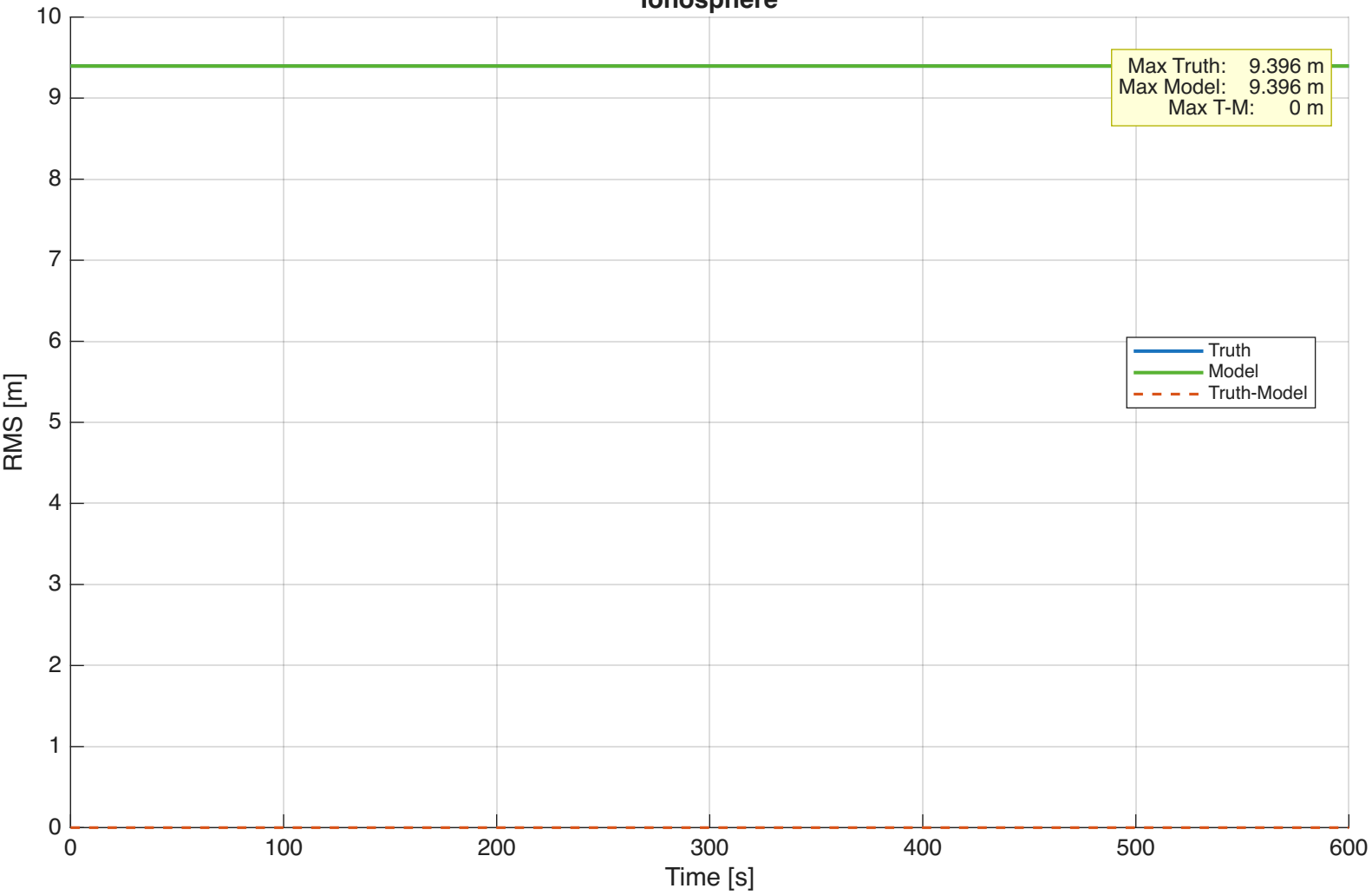
Code Noise



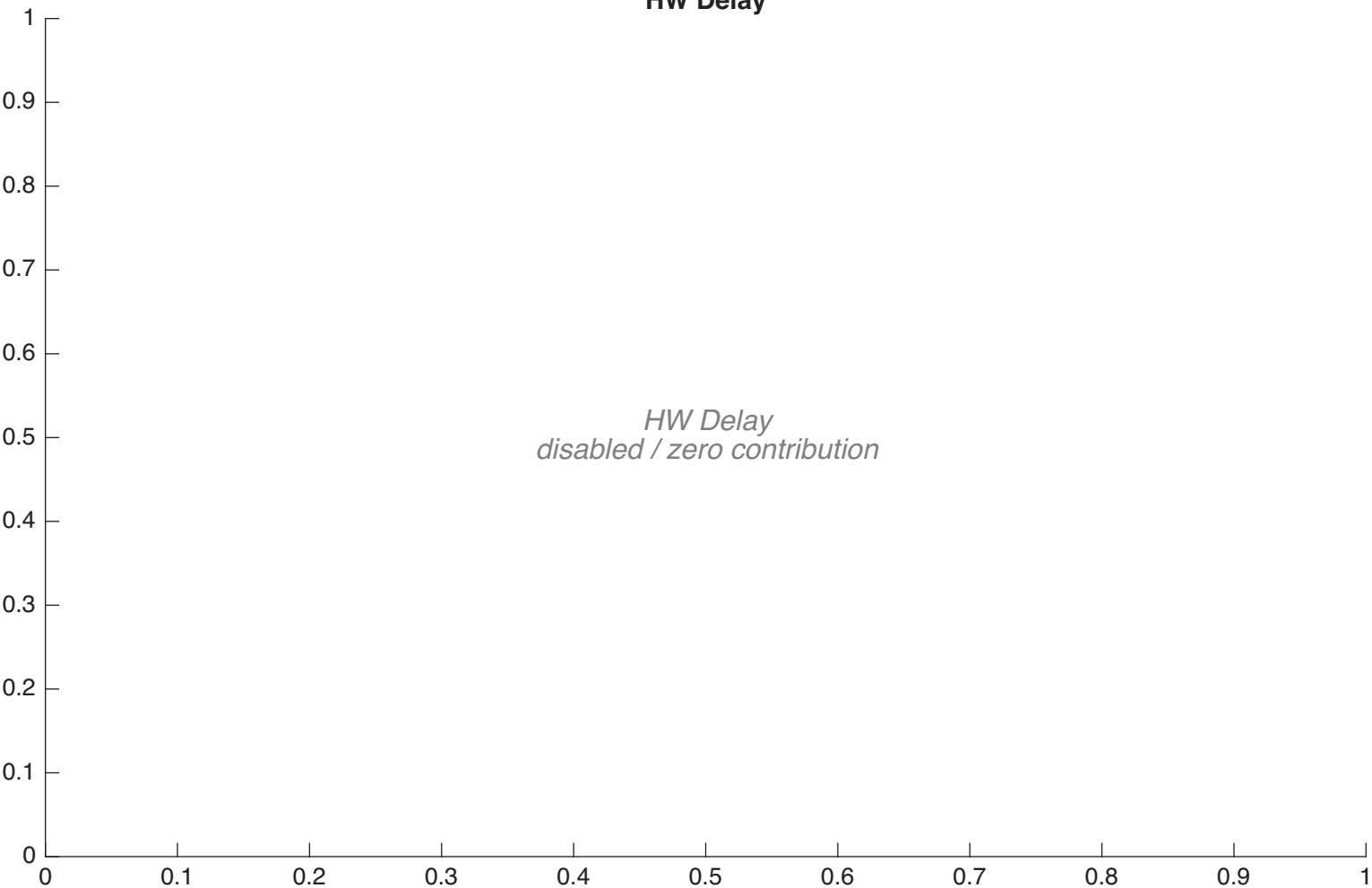
Troposphere



Ionosphere

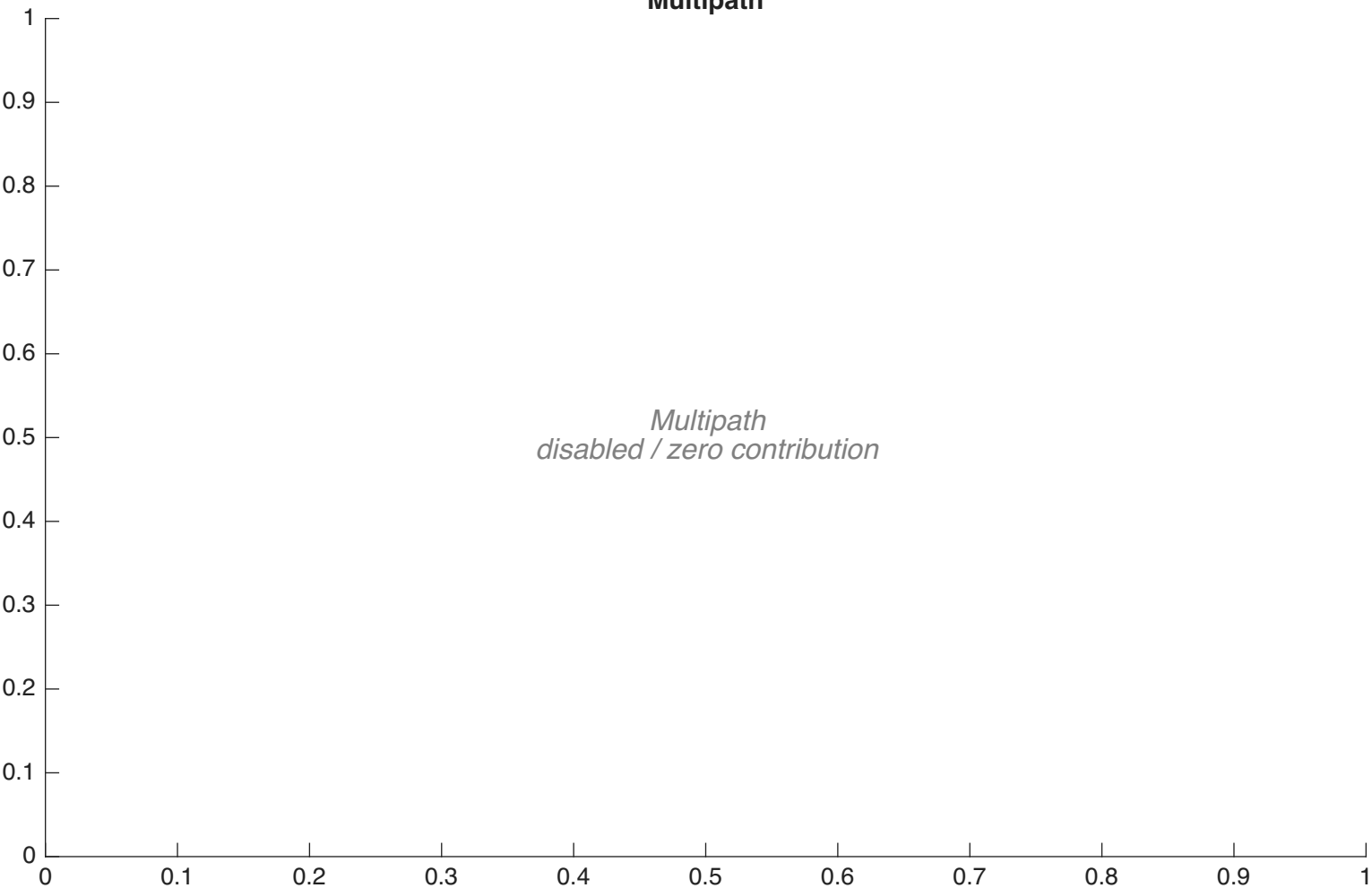


HW Delay

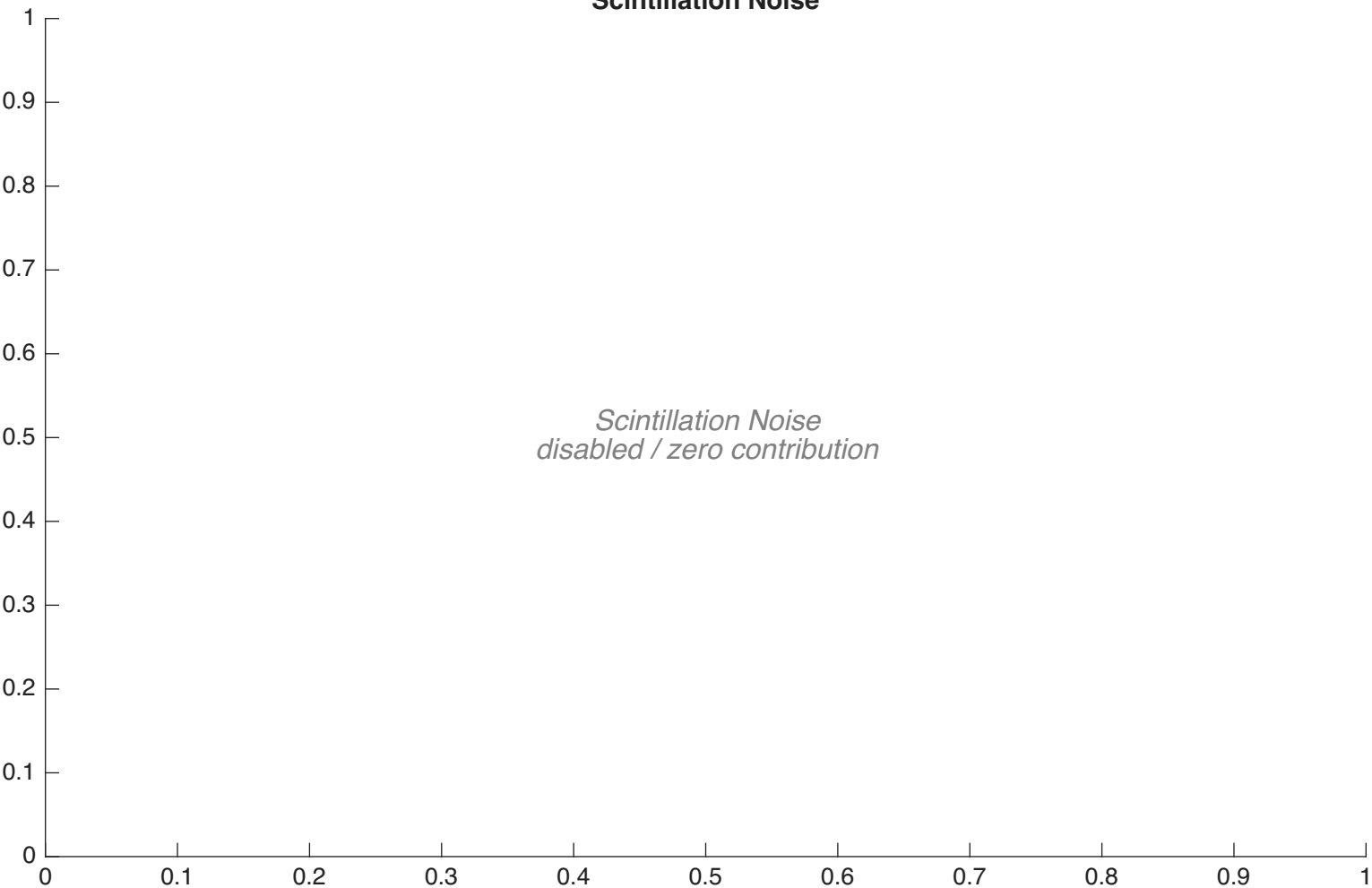


*HW Delay
disabled / zero contribution*

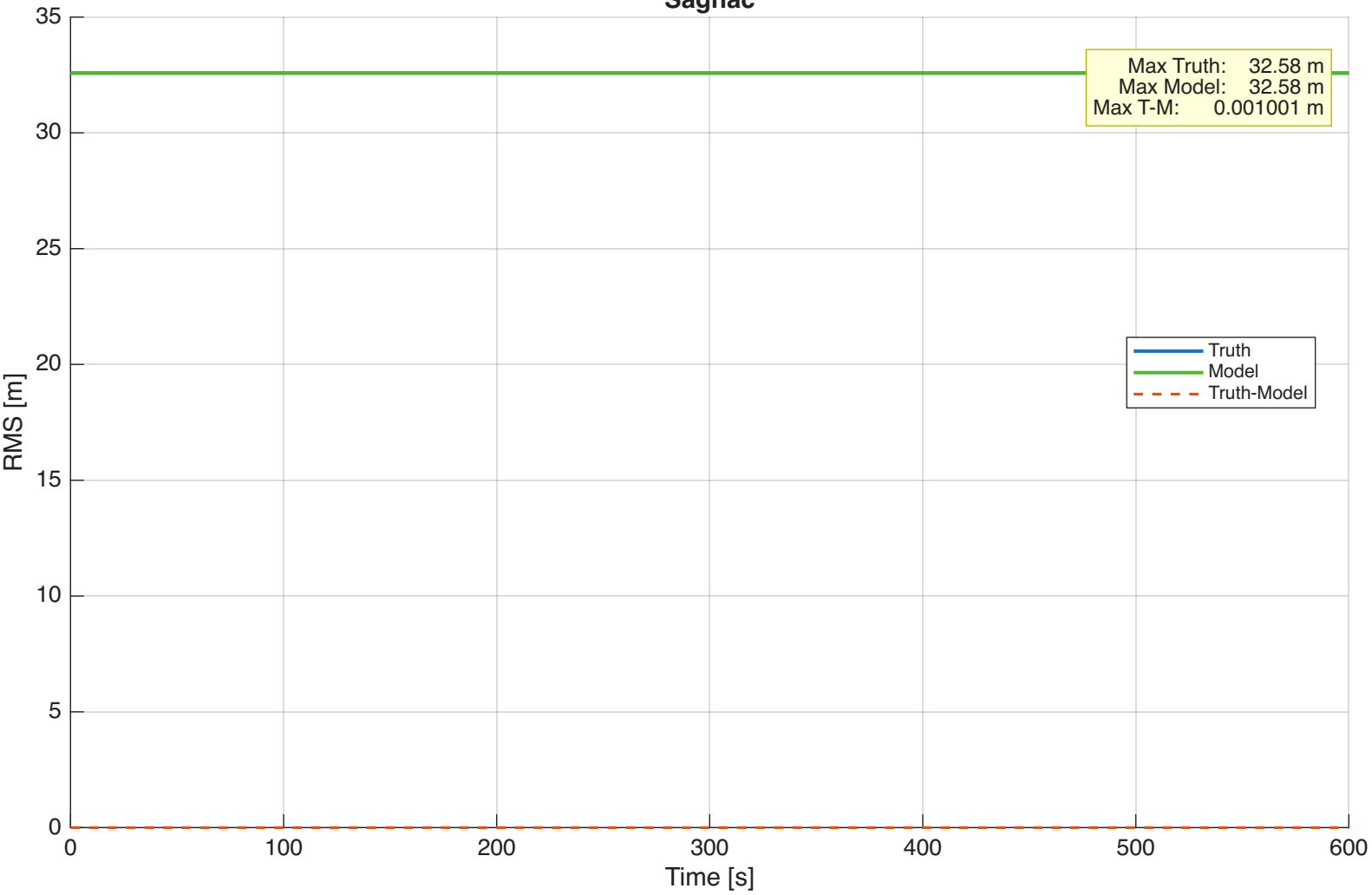
Multipath



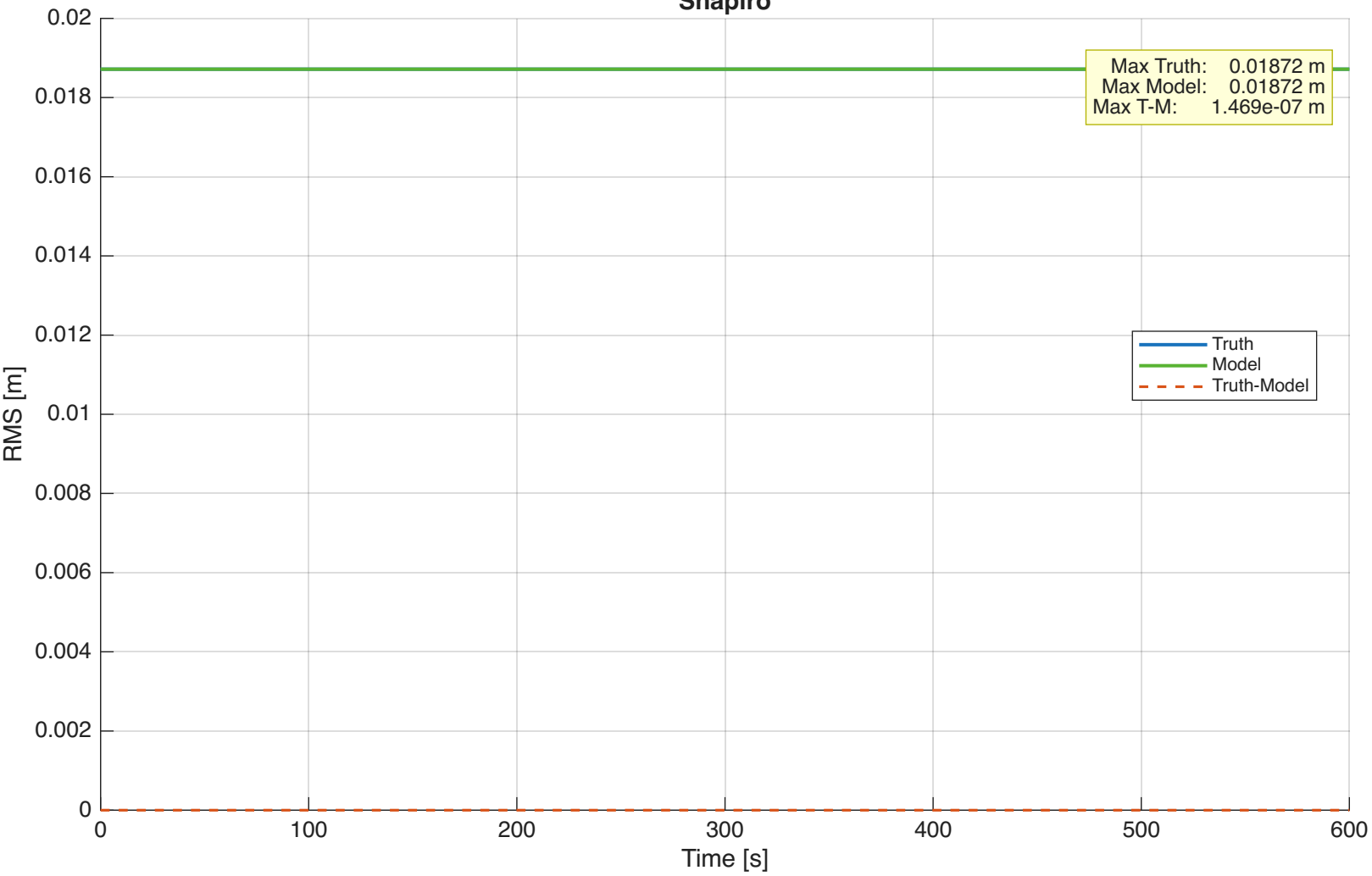
Scintillation Noise



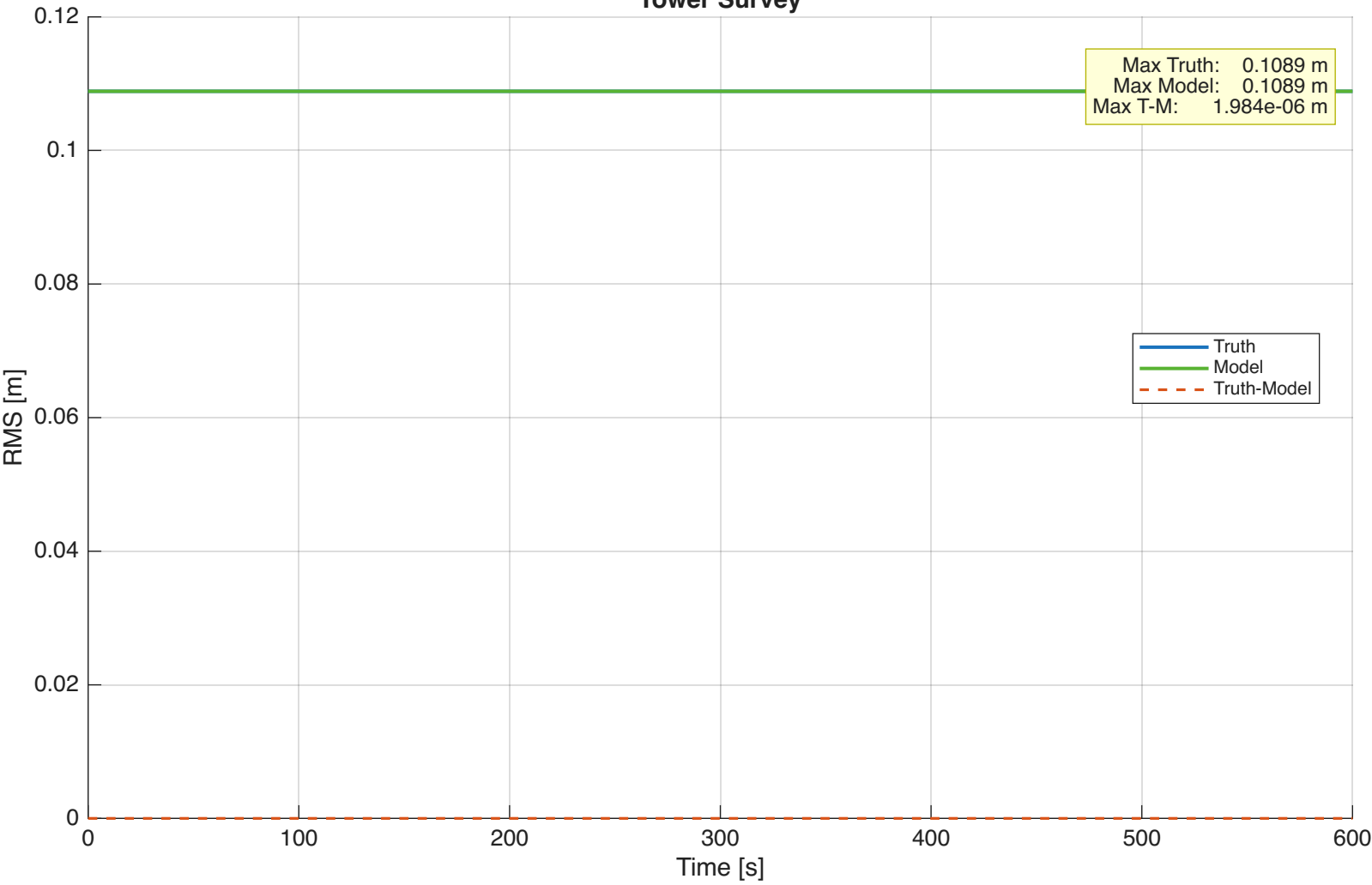
Sagnac



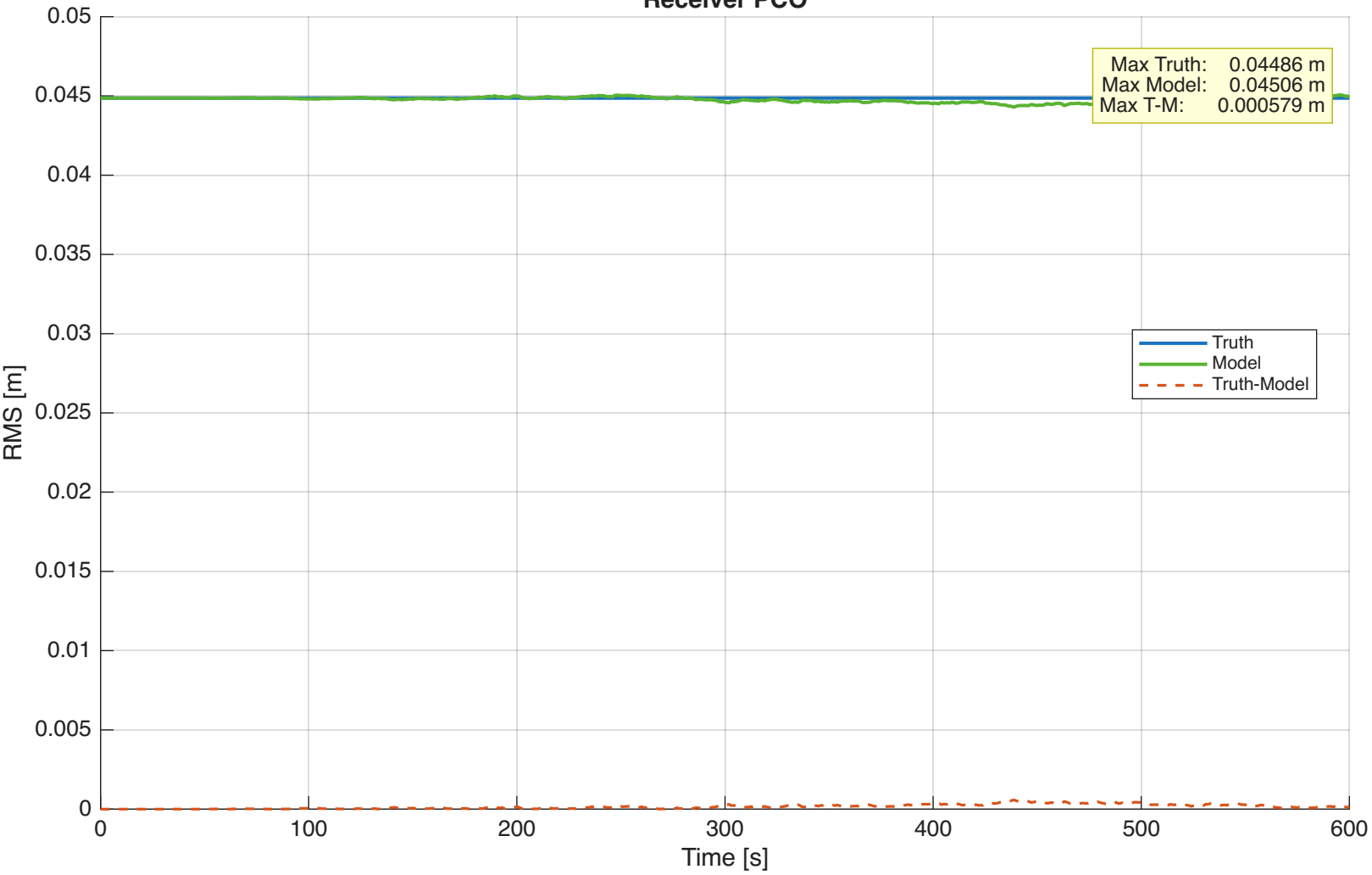
Shapiro



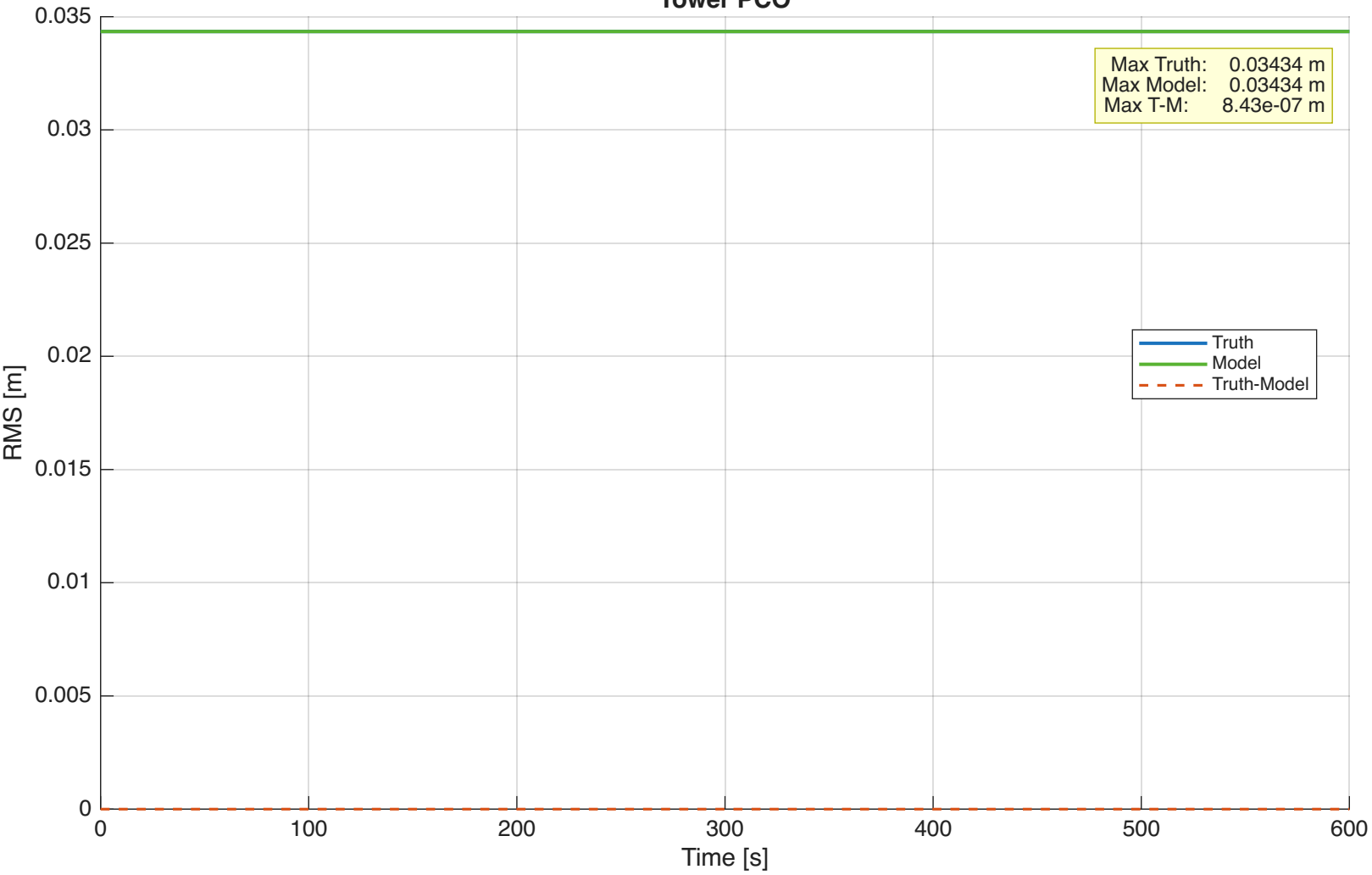
Tower Survey

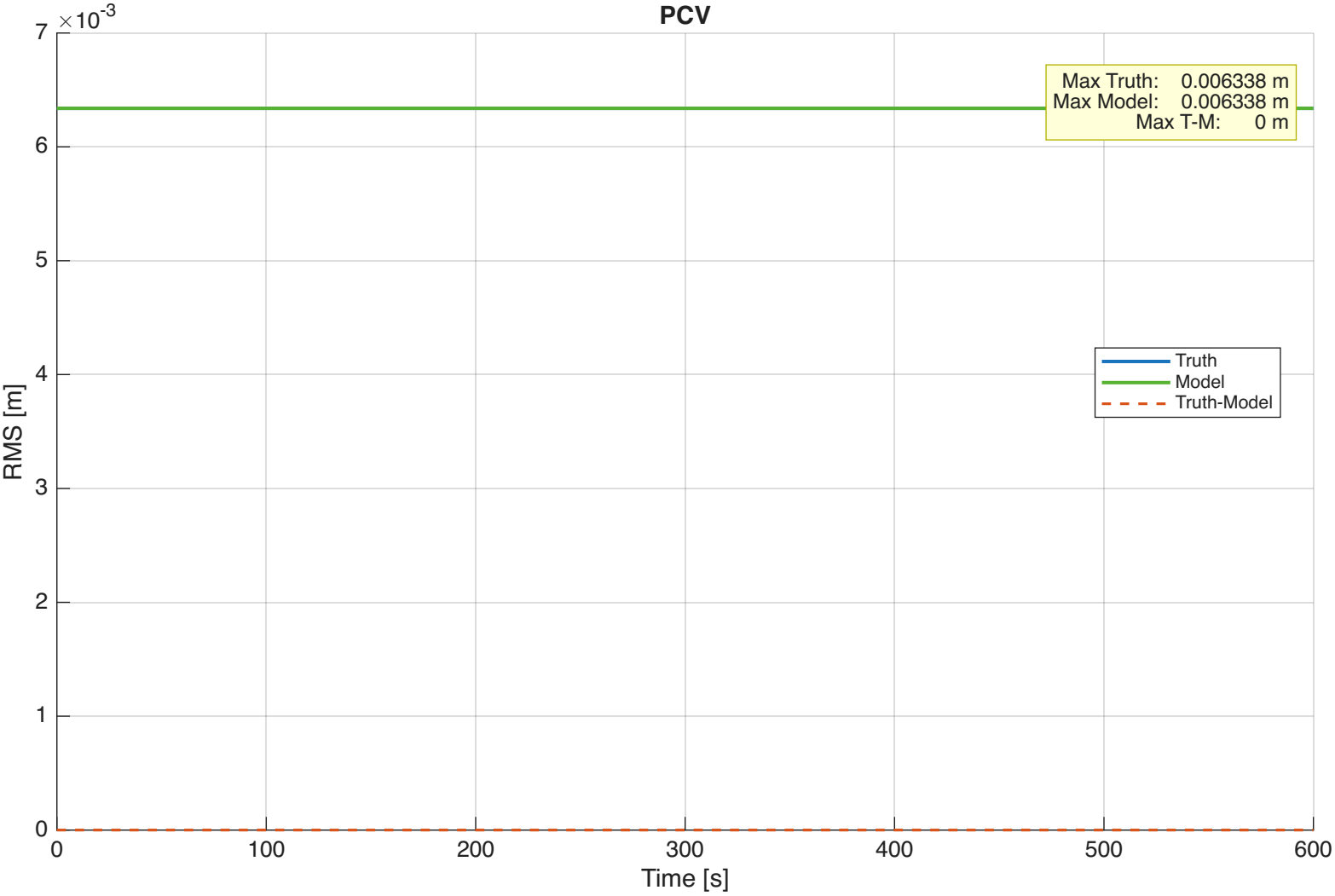


Receiver PCO

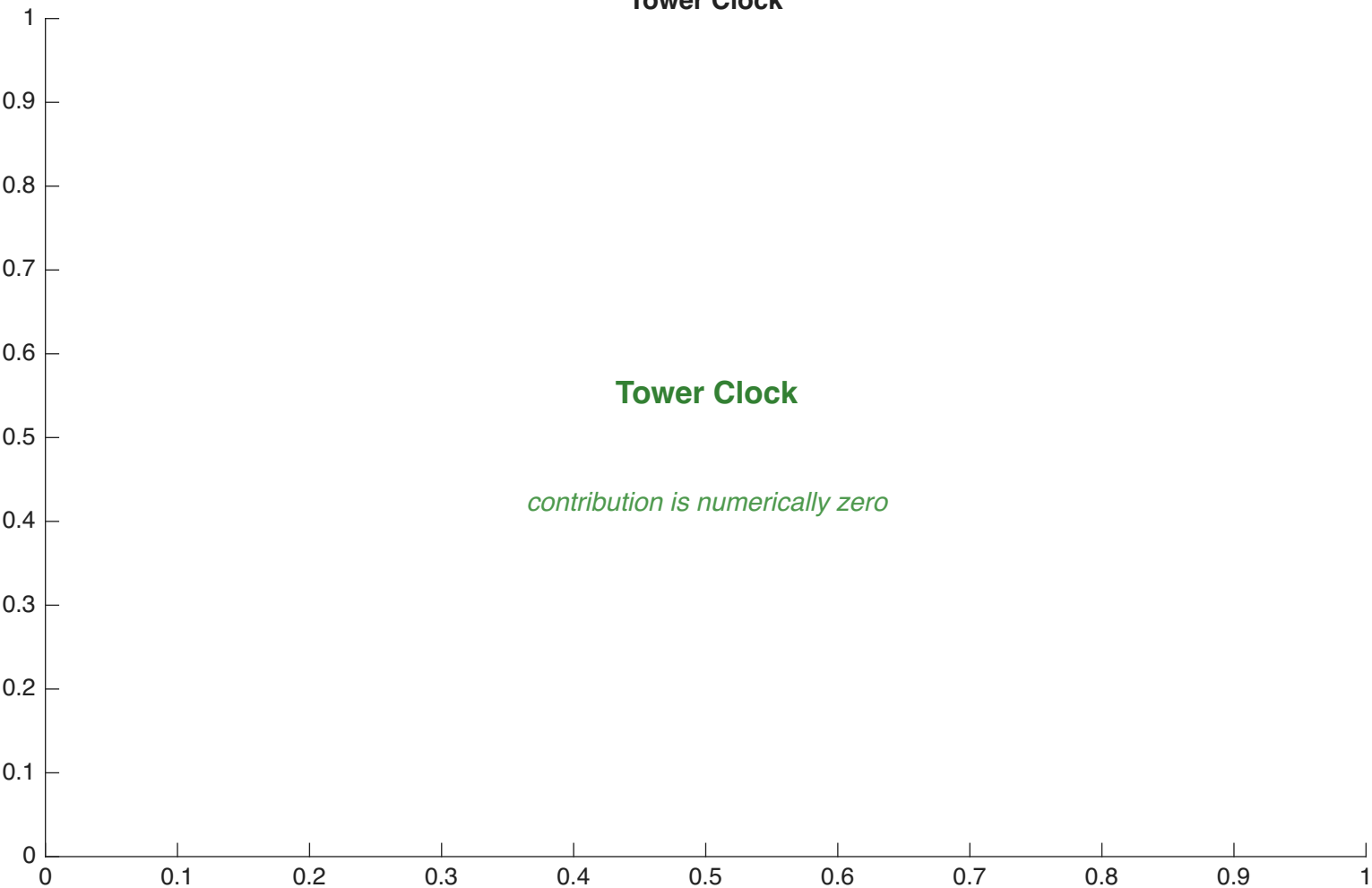


Tower PCO

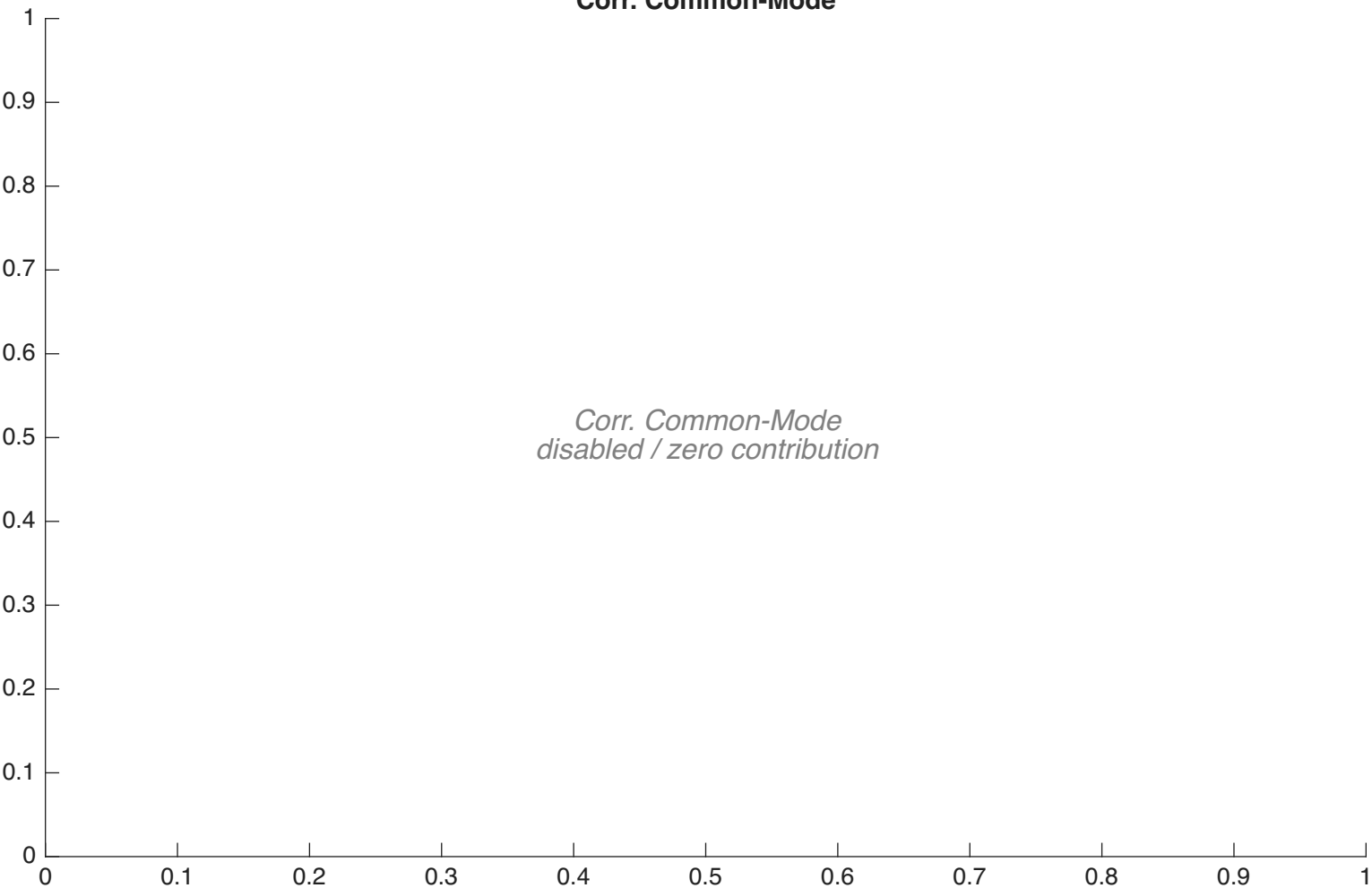




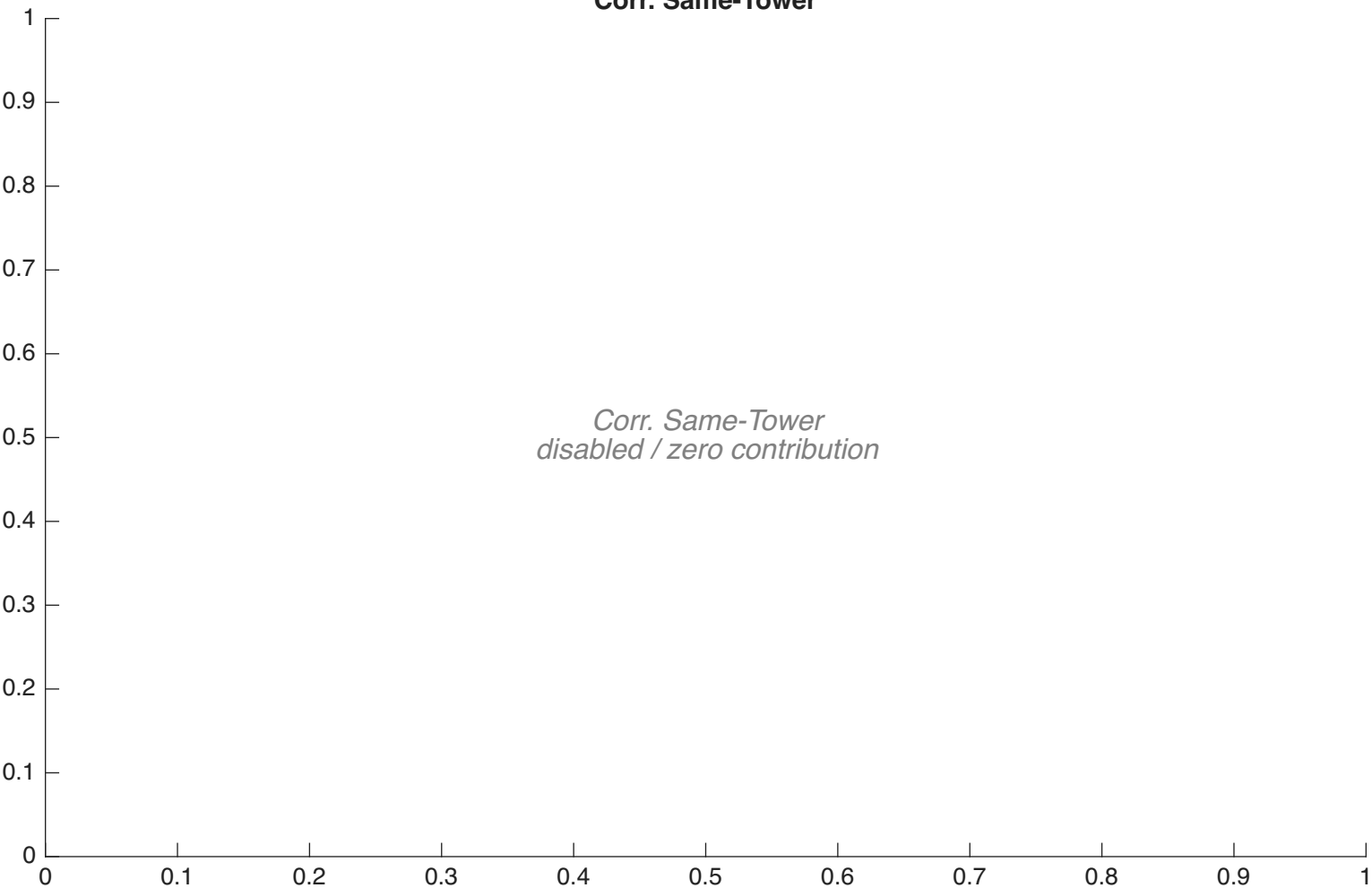
Tower Clock



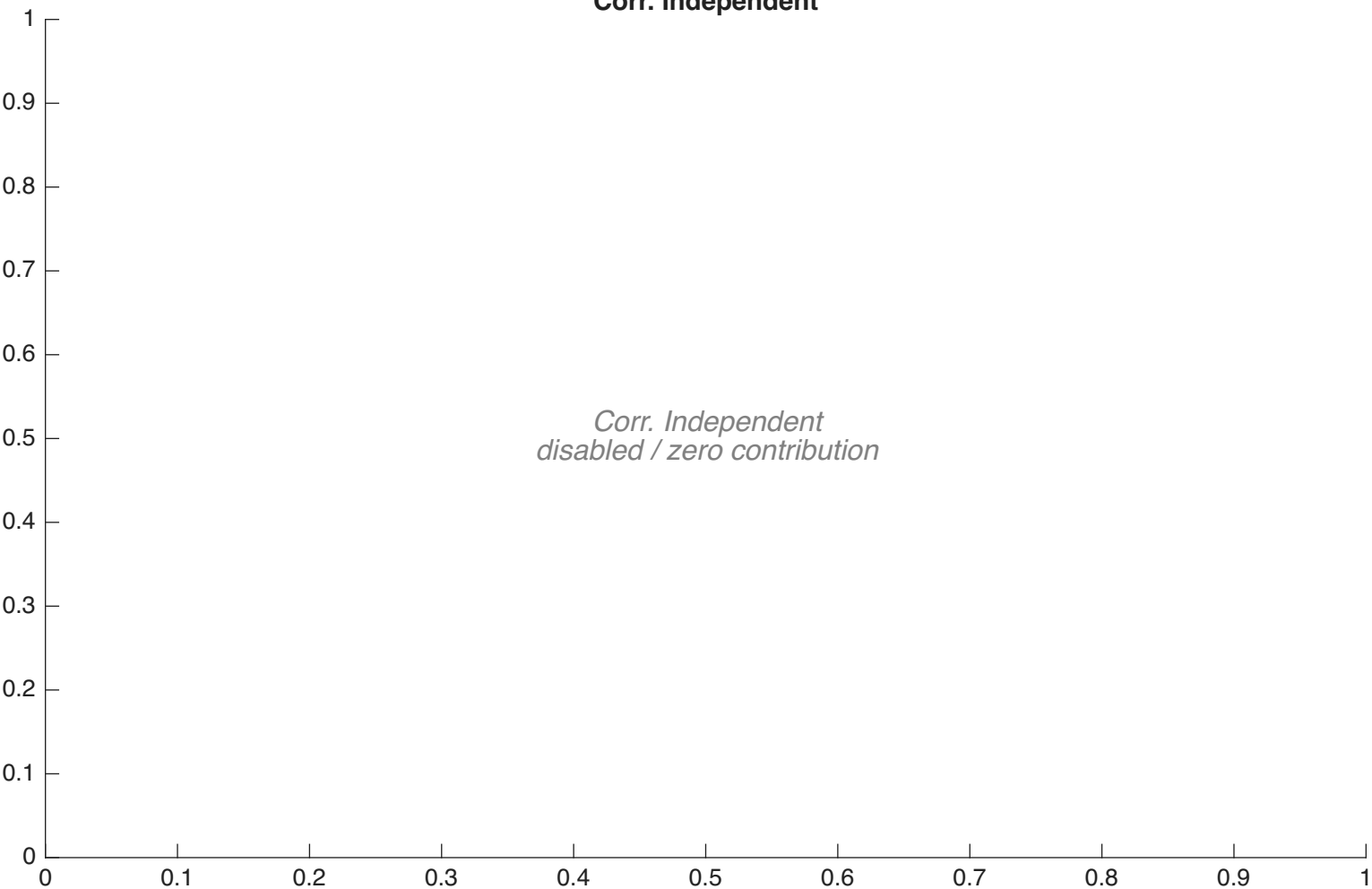
Corr. Common-Mode



Corr. Same-Tower

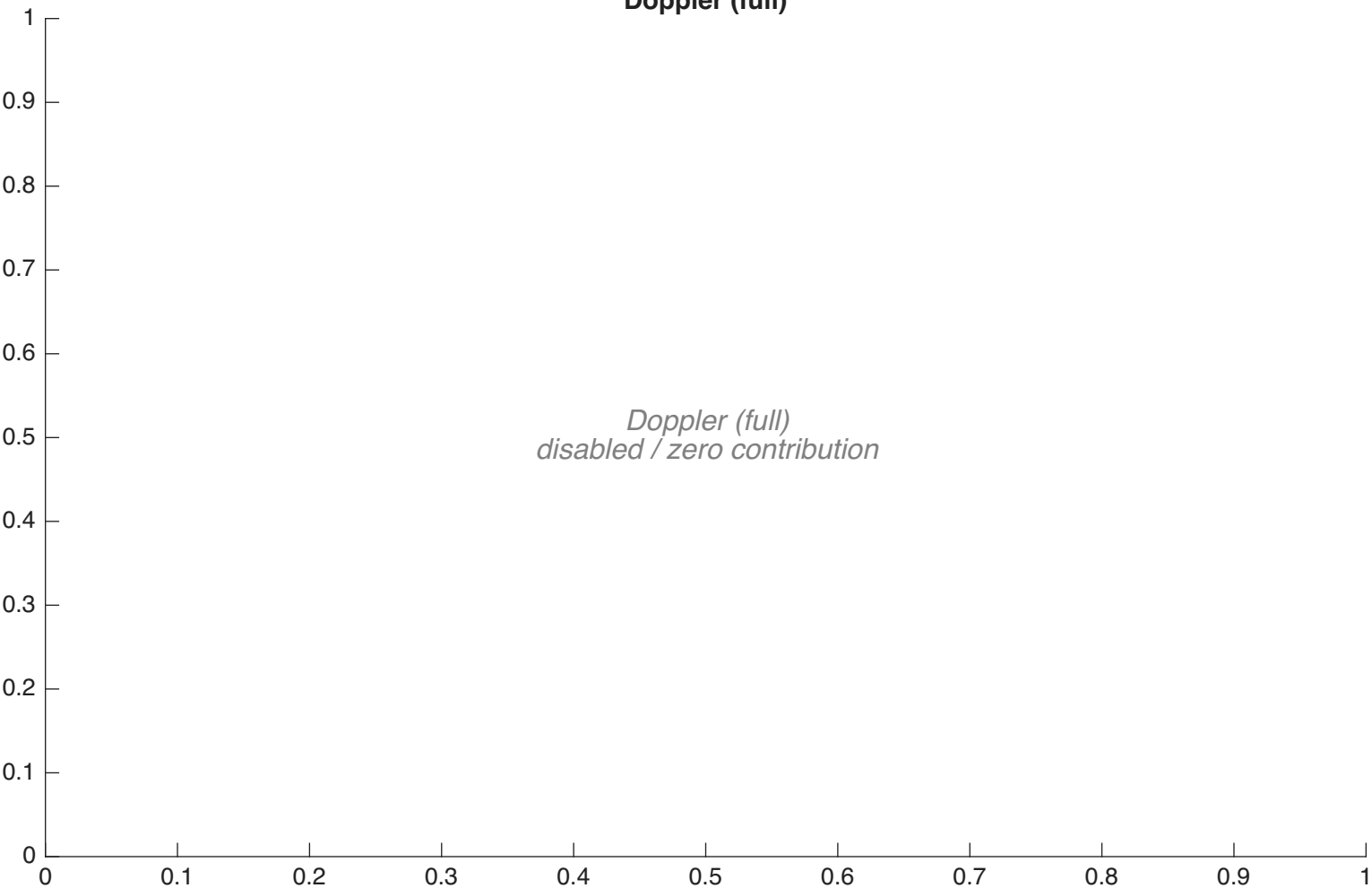


Corr. Independent

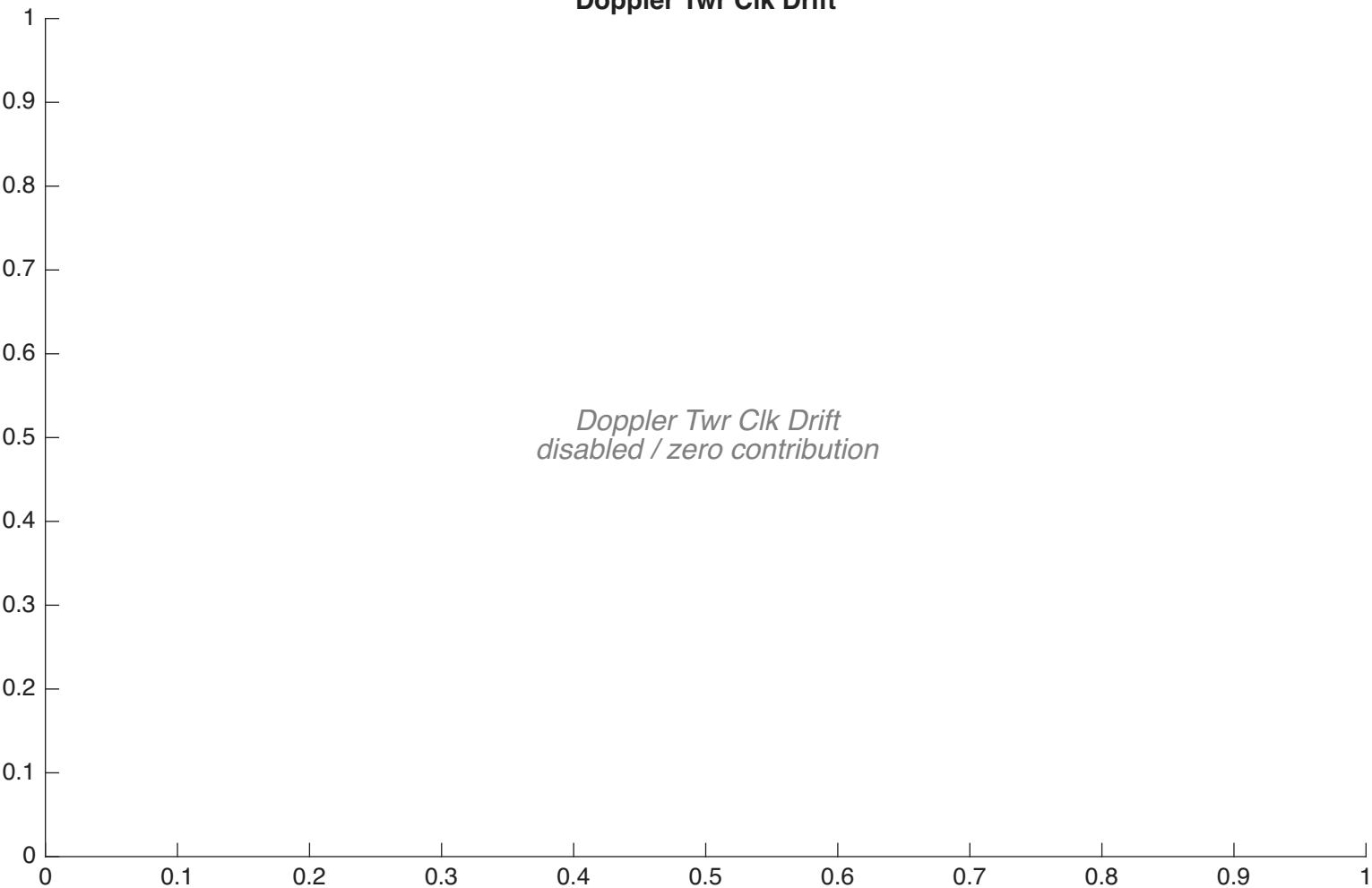


*Corr. Independent
disabled / zero contribution*

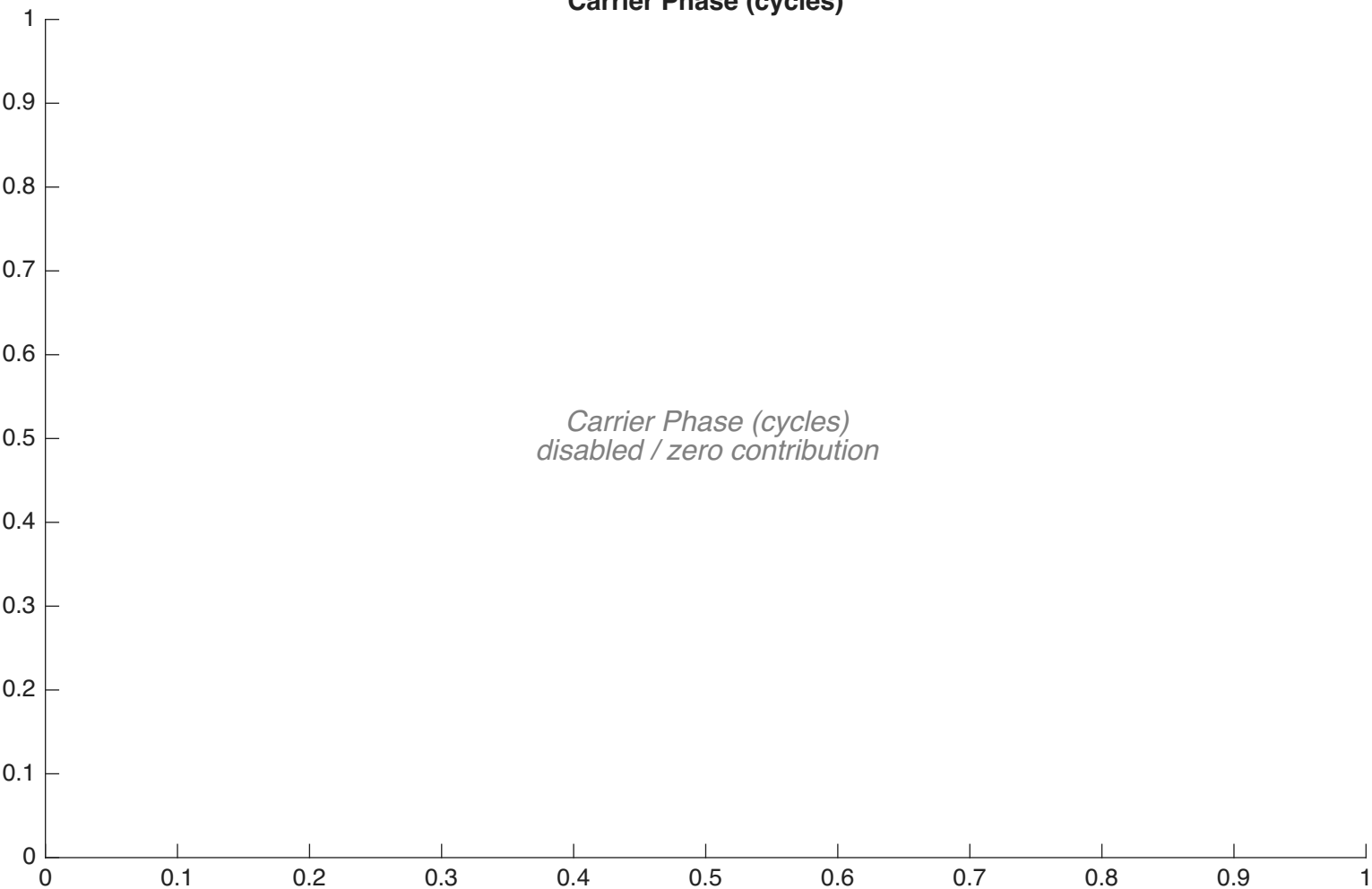
Doppler (full)



Doppler Twr Clk Drift



Carrier Phase (cycles)



Carrier Phase (m)

